

Refinements to Rceattle for eastern Bering Sea pollock

Rceattle 5.8.1 working paper for the 2024 assessment configuration

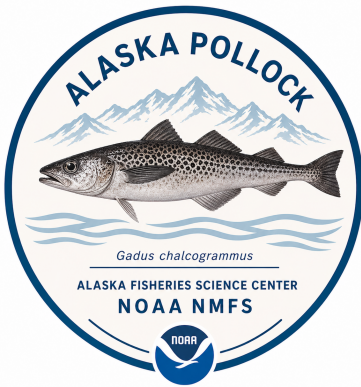
James Ianelli

Grant Adams

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Abstract

The custom AD Model Builder (ADMB) stock assessment model developed for eastern Bering Sea walleye pollock has a long history of application. For a variety of reasons, including general recommendations from the 2025 Center for Independent Experts (CIE) reviews, the custom ADMB code has been successfully ported to R Template Model Builder (RTMB). That conversion provides access to advanced RTMB features and makes the underlying code easier to understand and modify. Similarly, adopting more generic platforms, such as Rceattle, can provide access to more modern software developments. In this study, we tailored Rceattle code and configurations so that Rceattle can closely mimic the custom ADMB/RTMB assessment code. The Rceattle configuration uses the supported AFSC multinomial composition likelihood and a non-parametric selectivity form. We also evaluated a two-dimensional, first-order autoregressive (AR1) selectivity sensitivity; it did not converge and remains a research topic. An exploratory dynamic structural equation modeling (DSEM) analysis shows how recruitment covariates can be compared by cohort, separately from the final assessment configuration.



Rceattle WORKING PAPER

1 Executive Summary

Building from the ADMB-to-RTMB conversion, this analysis evaluates whether Rceattle can be configured to reproduce the main features of the 2024 eastern Bering Sea pollock “PM” assessment while providing a model that is easier to extend. This 5.8.1 update focuses on the Rceattle configuration, comparison with the modified ADMB model, model diagnostics, and projection inputs.

When the Rceattle parameters and fishery selectivity are set to the same values used in the ADMB model, Rceattle reproduces numbers-at-age, spawning biomass, and predicted catch to numerical precision. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. This result establishes agreement in the population dynamics when both implementations use the same parameter values.

When Rceattle was fit to the data, it reached OK convergence status with a positive-definite Hessian. The least-identified direction remained in annual ATS/AVO selectivity deviations, but the added direct coefficient-increment penalty reduced the condition number below the warning threshold. Relative to the ADMB fit, mean absolute differences were 1.4% for spawning biomass, 0.7% for recruitment, and 10.3% for total biomass. These small differences are concentrated in the earliest years; the remaining uncertainty is primarily estimability of the annual selectivity deviations.

The 2D age-by-year AR1 selectivity sensitivity was included as an initial diagnostic attempt. Results appeared broadly similar to the non-parametric constrained selectivity fits. However, because some parameters reached bounds, the model failed to converge. Closer examination of this selectivity form will continue as a research topic.

One-step-ahead residuals and nine terminal-year-consistent retrospective peels have been generated from the Rceattle configuration. Aggregate composition residual dispersion is lower than the standard-normal reference. Mohn's rho is 0.261 for spawning biomass, 0.119 for total biomass, 0.147 for recruitment, and -0.077 for fishing mortality. Forty-eight of 50 jitters returned to the best solution; two found a substantially poorer mode. These results focus subsequent work on selectivity identification, retrospective bias, and observation-model sensitivity.

2 Purpose and Scope

The September 2026 review focuses on whether the Rceattle configuration has sufficient documentation for Plan Team evaluation. This working paper serves that focused purpose rather than the broader purpose of a full Stock Assessment and Fishery Evaluation (SAFE) chapter.

This working paper therefore records:

- what was run from those development files;
- how the Rceattle outputs compare with the modified ADMB reference;
- which diagnostics are available now;
- which sensitivities remain research-only; and
- what work would help structure the next evaluation pass.

The report uses the [Alaska Stock Assessment Report \(asar\) package](#) for glossary conventions and accessibility guidance ([Schiano et al. 2026](#)). Those practices helped prepare the Quarto source for Section 508 review, including semantic headings and tables, descriptive figure captions, and alternative text. They support accessibility but do not replace final manual and agency-approved compliance checks.

Standard SAFE report sections such as background material, fishery history, ecosystem narrative, reference-point tables, and harvest advice are outside the scope of this document. Projection calculations are included only to test the export and use of the Rceattle results.

3 Rceattle Configuration

3.1 Model Inputs

The Rceattle configuration uses the prepared workbook `Data/EBS_24_pollock_m23_rceattle_full_1964-2024.xlsx`. The data builder retains every age from 1 through 15. Fishery compositions use nominal sample sizes; bottom trawl survey (BTS) and acoustic-trawl survey (ATS) sample sizes use the integer values aligned with the modified ADMB model. Age-1 survey selectivity remains separately parameterized from the age-2–15 curve, and each likelihood contribution evaluates the complete 15-age composition. Compositions use the supported `MultinomialAFSC` likelihood. We added a robustness constant of 0.001 to both observed and expected proportions before evaluating the likelihood. This prevents $\log(0)$ and limits the influence of observations in bins with expected proportions near zero.

The run uses Rceattle version 5.8.1 from upstream commit 7bb694788012, with the model configuration synchronized to `Rceattle-models` commit 3dc99ddfbf9e.

The next-assessment update script review is complete, and execution awaits complete new-year observations. The script duplicates terminal records and places placeholders for new catch, composition, and survey observations. The script serves as a template until complete new-year observations are available; after population, it creates a complete assessment workbook.

3.2 Selectivity

The configuration uses **NonParametricPM** fishery selectivity with annual deviations. When fit to the data, the model reached a small maximum gradient. The Hessian condition number was approximately 489,000 and received **OK** status. Its least-identified direction remained concentrated in **sel_coff_dev**, specifically the shared ATS/AVO block rather than the shared fishery/CPUE block.

The time-varying selectivity penalties use explicit standard deviations on log-selectivity random-walk increments. The fishery and its shared CPUE selectivity block use an SD of 0.500, ATS and AVO use 0.138, and BTS uses 1.000. ATS and AVO also use a direct coefficient-increment penalty (**Sel_curve_pen3 = 0.1**) to constrain the across-age common-mode direction that is removed by annual selectivity normalization. The fishery specification also retains the ADMB-matched curve-shape penalty weights (12.5, 1/60, and 1) and an average-selectivity centering weight of 10. These values are penalty coefficients rather than SDs.

The fitting sequence, composition-likelihood sensitivity, and 2D age-by-year AR1 development work are documented in the appendix.

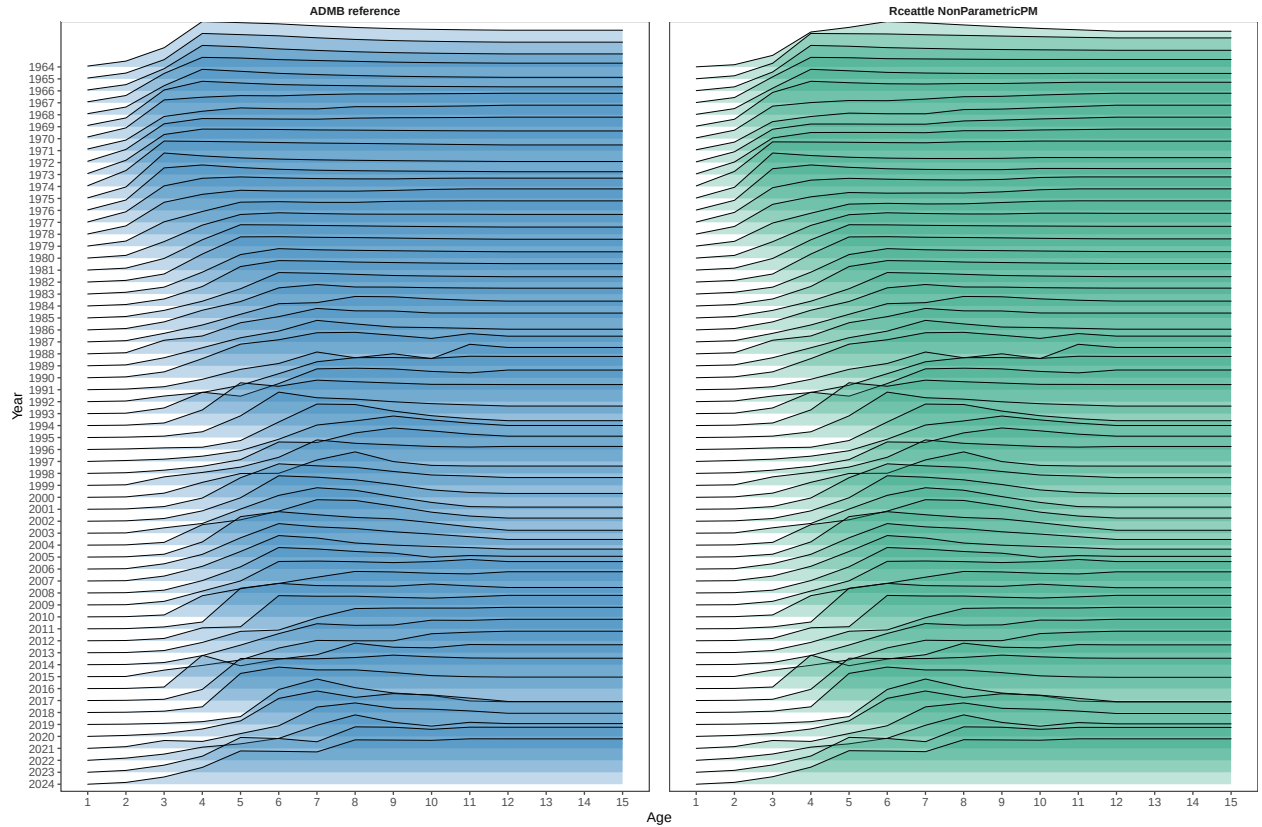


Figure 1: Relative fishery selectivity by year for the ADMB model and Rceattle configuration. Each annual curve is normalized to a maximum of one.

4 ADMB Comparison Results

When the Rceattle parameters and fishery selectivity are set to the ADMB values, the two programs give nearly identical population calculations. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. Annual Rceattle-to-ADMB ratios for both quantities remain extremely close to one (Figure 2).

The reported Rceattle likelihood contributions are 140.191 for indices, -123.761 for catch, 3572.874 for age compositions, and 98.379 for recruitment deviations. Their sum is 3704.588. The ADMB `tot_like` and `dat_like` values are 736.724 and 459.931, respectively, but cannot be compared directly because the two programs include different likelihood terms, penalties, and constants. The ADMB maximum absolute gradient is 3.47e-04; no Rceattle model-parameter gradient is available from this comparison because Rceattle does not estimate parameters in this run.

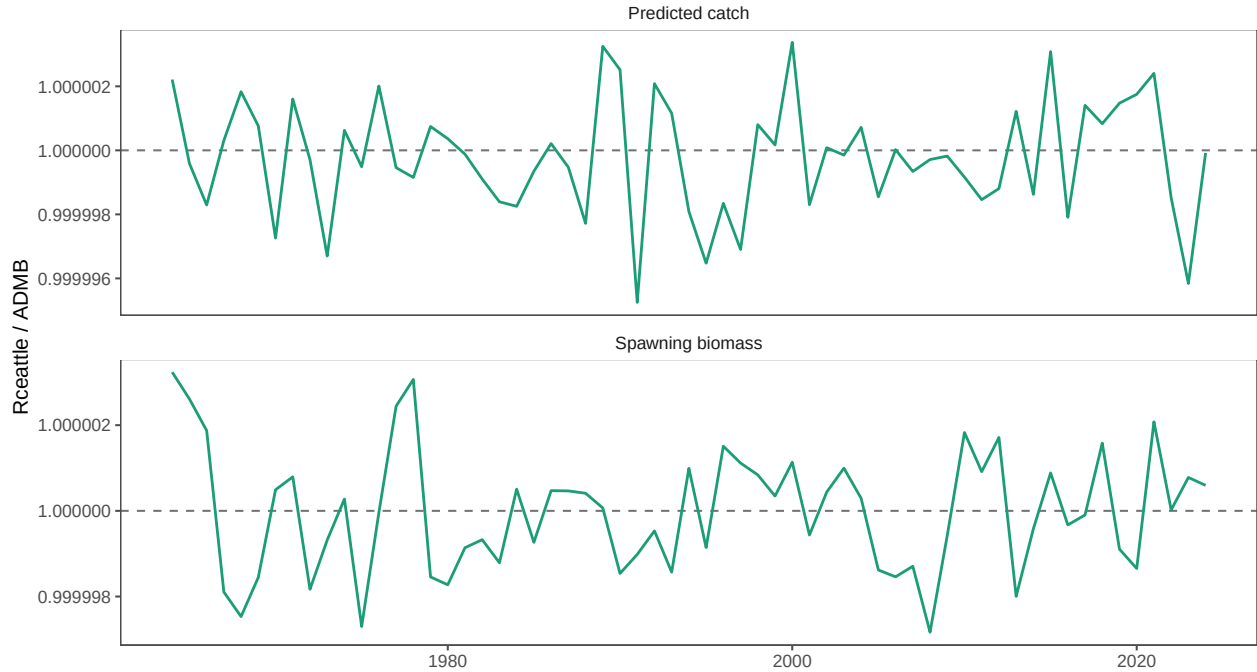


Figure 2: Rceattle-to-ADMB ratios for spawning biomass and predicted catch when Rceattle uses the ADMB parameter values.

Table 1: Agreement between fitted Rceattle trajectories and the modified ADMB reference. Correlation is the Pearson correlation across annual estimates. Mean absolute (%) and maximum absolute (%) are the average and largest absolute annual percentage differences, respectively. The 2024 difference (%) is the signed terminal-year difference of Rceattle relative to ADMB; negative values indicate that the Rceattle estimate is lower.

Quantity	Correlation	Mean absolute (%)	Maximum absolute (%)	2024 difference (%)
Rceattle configuration				
Recruitment	1.00	0.70	10.28	-6.45
Spawning biomass	1.00	1.39	10.01	0.06
Total biomass	0.98	10.32	95.63	2.61

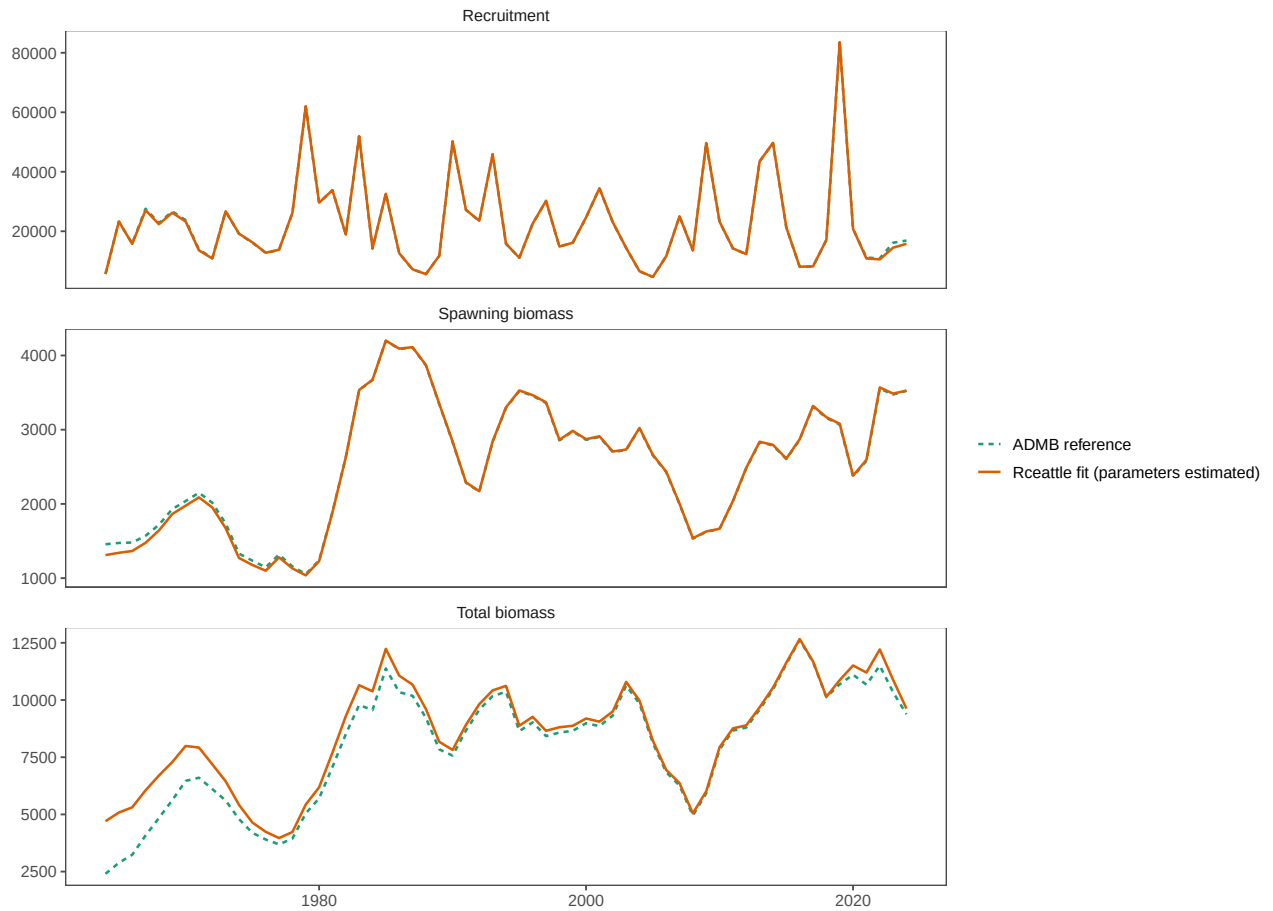


Figure 3: ADMB and fitted Rceattle trajectories for spawning biomass, recruitment, and total biomass. The Rceattle series is from the fit in which its parameters were estimated.

When Rceattle estimates its parameters, the fitted trajectories differ most from ADMB during the early period and in the terminal recruitment estimate (Figure 3). These trajectories are distinct from the run in which Rceattle parameters are set to the ADMB values. The fitted trajectories remain highly correlated with the ADMB reference, with relatively small mean annual differences (Table 1). Development sensitivities are documented in the appendix.

The Rceattle fit has a small maximum gradient and a positive-definite Hessian. Adding the direct ATS/AVO coefficient-increment penalty reduced the condition number from approximately 1.14 million to 489,000 and removed the prior Hessian-conditioning warning. The residual weakest direction remains almost entirely in the shared time-varying ATS/AVO selectivity coefficients, not the fishery/CPUE coefficients, so this block remains an appropriate focus for sensitivity checks even though the formal warning is resolved.

5 Diagnostics

The numerical convergence checks for the fitted Rceattle model, including the objective, maximum gradient, and Hessian status, indicated convergence (Table 2). The fit and status of the other diagnostic products are provided in Table 3.

Table 2: Stored Rceattle convergence checks for the evaluated configurations.

Model	Status	Objective	Max gradient	Weakest component	PD Hessian
Rceattle fit	OK	716.948070	0.000041	sel_inf	TRUE

Table 3: Diagnostic status for the September 2026 review.

Diagnostic	Current status	Review use
ADMB-parameter comparison Rceattle fit	Completed and saved in the ADMB-to-Rceattle comparison outputs. Completed with OK status, a positive-definite Hessian, and a condition number of approximately 489,000.	Primary numerical agreement check. Primary Rceattle comparison fit to the data.
2D AR1 selectivity sensitivity	Received a FAIL convergence status, an unsuccessful Hessian inversion, and parameter-bound warnings.	Requires further research before it can be considered as an alternative selectivity form.
OSA residuals	Completed for 1,864 residuals and saved in RDS outputs.	Useful, but analytical and exactly reproduced index groups need interpretation.
Rceattle retrospectives	Nine peels completed for the Rceattle fit with OK status.	The revised peels preserve data availability and fix terminal fishery/CPUE selectivity increments at zero, holding the terminal selectivity-at-age curve at its preceding-year value.
Legacy ADMB retrospectives	Existing ADMB retro.rep files are present.	Provides historical context distinct from the Rceattle retrospective peels.
Rceattle 5.8.1 self-tests	Fifty initial-start and 50 estimated-start refits completed with OK status.	Supports the revised positive-index simulation workflow; it does not resolve observed-data multimodality.
Jitters and likelihood profile	Forty-eight of 50 jitters returned to the best mode; two WARN fits reached an inferior mode. The age-3+ M profile was shallow with its grid minimum at 0.20.	Documents multimodality and weak information about natural mortality.

Table 4: Rceattle 5.8.1 self-test and paired-restart validation for the upstream-supported EBS configuration.

Metric	Value
Initial-start self-tests converged	5.000000e+01
Estimated-start self-tests converged	5.000000e+01
Paired fits with objective difference greater than 0.0001	0.00000e+00
Maximum absolute paired objective difference	3.09564e-08
Median across-simulation median SSB percent bias	-1.055%
Median terminal-year SSB percent bias	-0.907%
Fleets with at least one row above 2 percent rejection probability	2.00000e+00
Maximum row-level non-positive probability	6.66814e-02

Table 5: Fleet-level probability of a non-positive index draw under the fitted, untruncated observation distribution.

Fleet	Code	Distribution	Rows	Maximum row probability	Mean row probability	Rows above 2
ATS	4	Lognormal	19	0.00%	0.00%	
ATS_1	6	Lognormal	18	0.00%	0.00%	
AVO	2	Normal	18	3.88%	0.23%	
BTS	3	MVN	42	6.64%	0.17%	
BTS_1	5	Lognormal	42	0.00%	0.00%	
CPUE	7	Normal	12	0.00%	0.00%	

5.1 Rceattle 5.8.1 self-test validation

Rceattle 5.8.1 changed simulation of natural-scale normal and multivariate normal survey indices. The fitted likelihood remains an untruncated normal; the simulator redraws non-positive values because `data_check()` excludes them. The resulting self-test therefore uses a normal distribution truncated at zero whenever rejection occurs.

The ADMB-to-Rceattle comparison was rerun first with Rceattle 5.8.1. Its saved summary gives mean absolute differences of 0.000117% for spawning biomass and 0.000143% for predicted catch. This confirms that the package update preserved the agreement in the population calculations.

The self-test uses the same delivered EBS workbook and supported `MultinomialAFSC` composition criterion as the Rceattle configuration. The observed-data fit reproduced the stored objective of 716.948070, maximum gradient of 0.000041, and OK status with a positive-definite Hessian. The Hessian condition number was approximately 4.9×10^5 ; its least-identified direction remained concentrated in the shared ATS/AVO `sel_coff_dev` block rather than fishery/CPUE.

All 50 phased simulations converged with OK status when refitted from the original initial values. Repeating the same simulated datasets from the fitted parameter values also produced 50 OK fits. Paired objectives agreed within 3.1×10^{-8} , so these simulations showed no start-dependent mode. Median spawning-biomass bias was -1.06% across years and -0.91% in the terminal year. These results validate the revised self-test workflow; they do not rule out the mode previously encountered when fitting the observed data.

One AVO row had a 3.76% non-positive probability and one BTS multivariate-normal row had a 6.67% probability. Those rows exceed the package's 2% warning threshold and should be identified when interpreting recovery. CPUE and the remaining index series had negligible or zero marginal rejection probability.

5.2 OSA Residuals

OSA residuals run on the Rceattle configuration. The aggregate composition fits are generally close, while all three OSA quantile plots are narrower than the standard-normal reference (Figure 4). These diagnostics apply to the fitted `NonParametricPM` selectivity configuration. The failed 2D-AR1 sensitivity and ADMB-to-Rceattle calculation use separate results. The composition diagnostics use `afscOSA` version 0.0.1 and pass Rceattle’s internally calculated residuals to `afscOSA::run_osa()`. The residual builder uses complete ages 1–15, with nominal fishery sample sizes and the integer BTS and ATS sample sizes aligned with the modified ADMB model. This follows the current [afscOSA interpretation guidance](#) while retaining Rceattle’s residual calculation.

Following that guidance, the aggregate age-composition fits are the first check for systematic misfit, including patterns that could indicate an unsuitable selectivity form. The red vertical ranges contain 95% of data simulated from the fitted multinomial model conditional on the aggregate sample size and estimated parameter values held constant. They represent conditional simulation variability. The quantile–quantile plots then assess whether the OSA residuals resemble a standard normal distribution. The standard deviation of normalized residuals (SDNR) should be near one, and the displayed tail statistics provide a separate check on extreme residual behavior. Finally, the OSA and Pearson bubble plots are used to look for age, year, or cohort patterns that may suggest the source of misfit. Each statistic and nominal interval contributes to an integrated diagnostic interpretation rather than an automatic model-rejection rule.

The aggregate observed and fitted compositions are generally close, although the survey panels contain some age-specific departures. The OSA distributions are narrower than the standard-normal reference for all three composition sources: SDNR is 0.64 for the fishery, 0.80 for BTS, and 0.53 for ATS. Their tail diagnostics also show shorter-than-expected tails. Together these results indicate underdispersion relative to the fitted multinomial observation model. Model selection requires this information alongside the full suite of diagnostics. The bubble plots should instead be used to identify persistent age, year, or cohort patterns and to guide focused checks of selectivity, composition weighting, and observation-model assumptions. One Pearson residual (8.86) is capped at 6 in the figure, following the package’s plotting convention.

5.3 Retrospective patterns

Nine retrospective peels were run from the Rceattle fit. The unpeeled model ends in 2024, and the peeled fits end in 2023 through 2015. Each peel preserves the data-availability pattern of the 2024 model. Thus, fishery age compositions end one year before the peel terminal year, survey series retain their historical sampling schedule and terminal lags, and earlier discontinued series retain their original ending years.

Each peel applies the fitting sequence documented in the appendix. The terminal-year fishery and mirrored catch-per-unit-effort selectivity increments are fixed at zero. All nine final fits received OK convergence status, with maximum gradients from 2.1e-05 to 6.4e-05.

The `NonParametricPM` fishery selectivity is parameterized at age. The base `sel_coff` parameters and annual `sel_coff_dev` random-walk increments apply to age bins; the fishery uses 12 estimated bins and holds older ages at the last estimated value. Fixing every terminal-year increment at zero therefore holds the realized terminal selectivity-at-age curve at the preceding-year curve. Across all peels and both the fishery and mirrored catch-per-unit-effort series, the maximum absolute terminal-to-preceding-year difference was 8.9e-16. These Rceattle fits are separate from the legacy AD Model Builder (ADMB) retrospective files.

The estimated Mohn’s rho values are 0.261 for spawning biomass, 0.119 for total biomass, 0.147 for recruitment, and -0.077 for fishing mortality. The positive biomass and recruitment values indicate that the peeled models generally estimate higher quantities at their terminal years than the corresponding unpeeled estimates. The negative fishing-mortality value indicates lower terminal fishing mortality in the peels. Spawning biomass shows the largest average relative difference. The consistent OK statuses and small gradients support interpretation of these values as properties of the revised retrospective specification, while the magnitude of spawning-biomass rho motivates continued evaluation of selectivity, survey influence, and recent recruitment estimates.

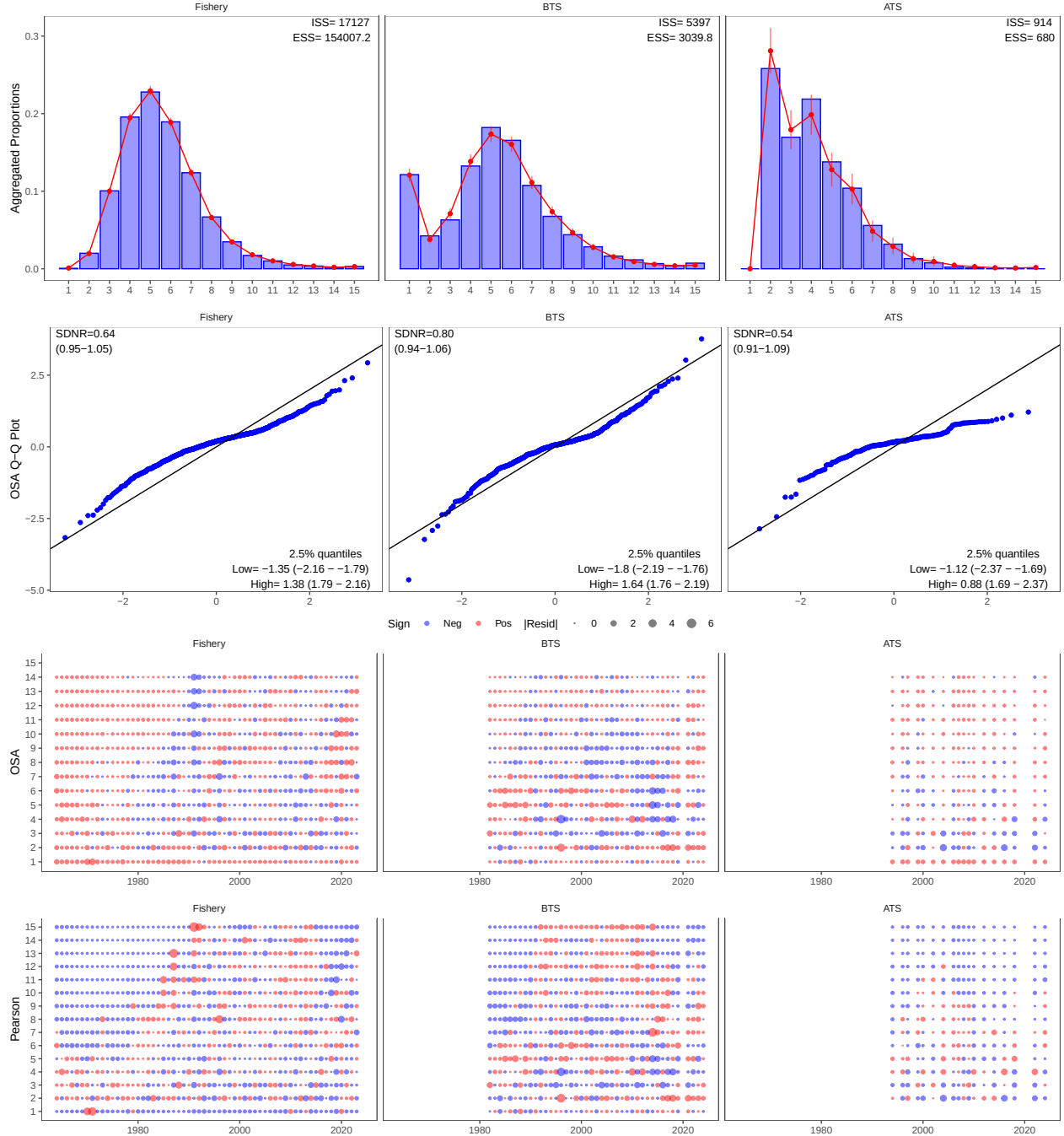


Figure 4: Age-composition diagnostics for the Rceattle fit. The fit uses complete ages 1–15, nominal fishery sample sizes, and BTS and ATS integer sample sizes aligned with the modified ADMB model. Diagnostics were produced with afscOSA for the fishery, bottom trawl survey (BTS), and acoustic-trawl survey (ATS). The top row compares aggregate observed proportions (blue bars) with fitted proportions (red points and lines); red vertical ranges contain 95% of data simulated conditionally from the fitted multinomial model, aggregate sample size, and estimated parameter values held constant. The second row shows quantile–quantile plots of Rceattle’s internal one-step-ahead (OSA) residuals with the standard deviation of normalized residuals (SDNR), its expected 95% interval, and lower and upper tail diagnostics. The lower rows show OSA and Pearson residuals by age and year; red and blue indicate positive and negative residuals, and bubble area represents magnitude. Absolute residuals greater than 6 are capped at 6 for display. The fitted model retains ATS 2020; its sample size of one yields zero rounded counts in the afscOSA plotting conversion, so the plotting layer omits that row.

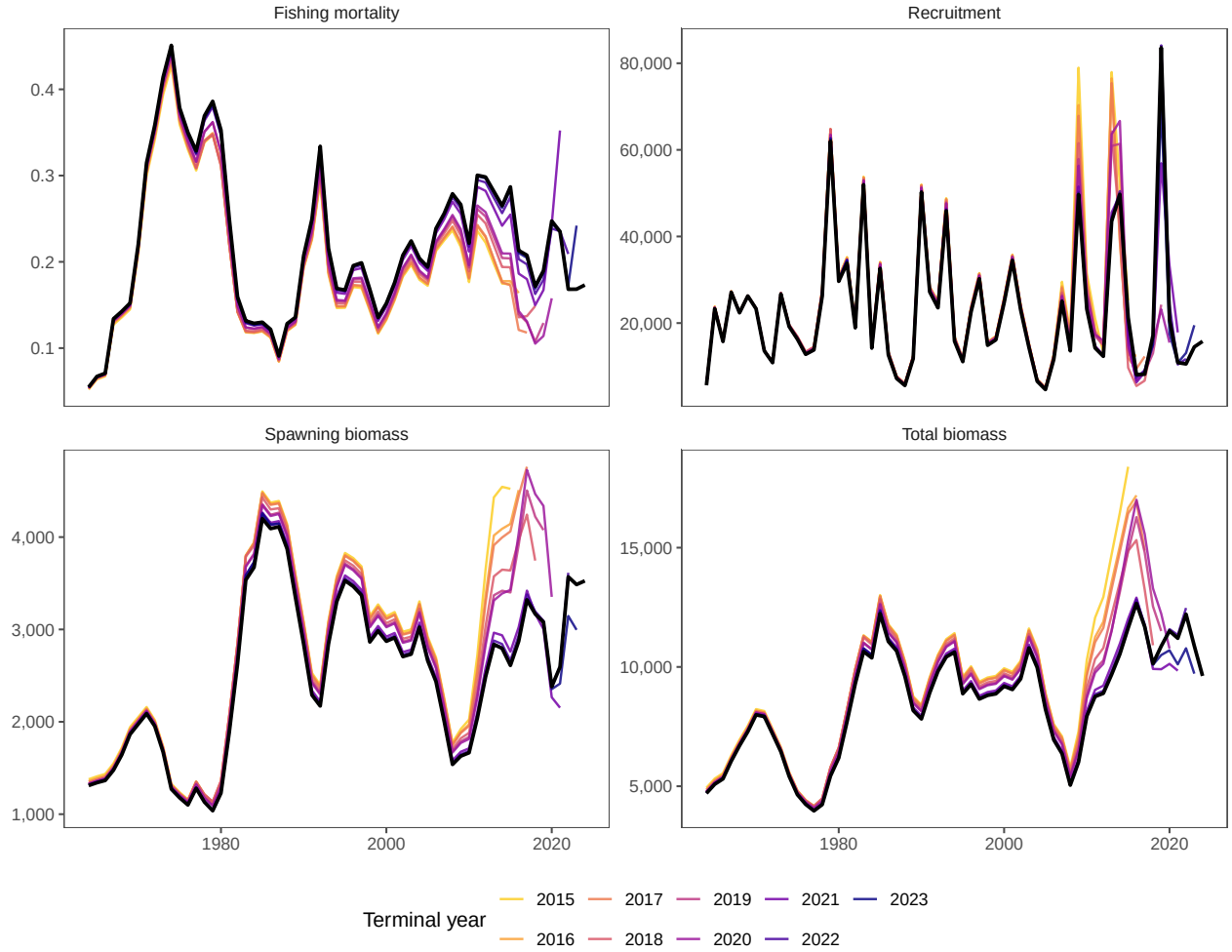


Figure 5: Nine retrospective peels for the Rceattle configuration. The black line is the unpeeled model ending in 2024; colored lines are successive peels ending in 2023 through 2015 and are plotted through each peel’s terminal year. Each peel preserves the original data-availability lags and fixes the terminal fishery/CPUE selectivity-at-age increment at zero, holding the terminal curve at its preceding-year value. All nine fits received OK convergence status.

Table 6: Mohn’s rho for the nine retrospective peels of the Rceattle configuration. Values are the mean relative difference between a peeled estimate in its terminal year and the corresponding estimate from the unpeeled model. Each peel preserves the 2024 data-availability pattern and fixes the terminal fishery/CPUE selectivity-at-age increment at zero.

Quantity	Peels	Mohn’s rho
Total biomass	9	0.119
Spawning biomass	9	0.261
Recruitment	9	0.147
Fishing mortality	9	-0.077

6 Projections

The saved Rceattle fit supplies every population and fishery input for the Standard Projection Model calculations implemented in `spmR` (J. Ianelli 2026). The run uses `spmR` version 0.3.0 from pinned main-branch commit `e86fc6aa`. The adapter in `R/write_spmr_projection_inputs.R` writes terminal abundance, natural mortality, maturity, spawning and fishery weights, the arithmetic mean of fishery selectivity-at-age over the most recent five assessment years (2020–2024), recruitment, and spawning-biomass histories to `pm.prj`. The same five years define the recent average fishing mortality. The reproducible runner `R/run_spmr_tier3_projection.R` then applies those inputs to all seven Tier 3 alternatives. The report reads the saved projection products, so every table and figure in this section comes from the same saved Rceattle fit. The complete field mapping and validation sequence are documented in Appendix A-2.

Table 7: Projection files generated from the saved Rceattle fit for the `spmR`/SPM calculations. Checksums show the first eight MD5 characters; the complete values are retained in `manifest.csv`.

spmR Projection Files

file	role	exists	checksum
<code>spm.dat</code>	SPM setup	TRUE	8b4f5e91
<code>pm.prj</code>	Rceattle-derived species input	TRUE	e91692f1
<code>tacpar.dat</code>	TAC parameters	TRUE	3b02746e
<code>spm</code>	SPM executable	TRUE	6f52316e
<code>age_schedules.csv</code>	Rceattle-derived age schedules	TRUE	3c450f47
<code>fishery_selectivity_recent_years.csv</code>	Annual fishery selectivity used in the recent-year average	TRUE	0377c30e

Table 8: Key settings and source information for the Rceattle-derived SPM projection.

Generated SPM Setup and Source Information

Item	Value
Rceattle source fit	<code>nonparametric_pm</code>
Source checksum	334f8c14b918536c53e4abaa0fc116ae
Composition likelihood	MultinomialAFSC
Composition sample sizes	fishery nominal; BTS and ATS integer sample sizes aligned with modified ADMB
<code>spmR</code> version	0.3.0
<code>spmR</code> commit	<code>e86fc6aa</code>
Terminal assessment year	2024
Fishery selectivity rule	arithmetic mean of annual selectivity-at-age
Fishery selectivity years	2020, 2021, 2022, 2023, 2024
Recent average-F years	2020, 2021, 2022, 2023, 2024
Spawning timing	0.25 year from January 1 (start of April)
Rceattle encoding	<code>spawn_month</code> = 3 elapsed months; $3/12 = 0.25$
SPM encoding	<code>Spawnmo</code> = 4 as a one-based calendar-month code; $(4 - 1)/12 = 0.25$
Timing compatibility check	Matched: both calculations apply survival through 0.25 of the year
Spawning-biomass timing convention	Mature spawning biomass after survival through 0.25 of the year: $N \times \exp(-0.25 Z) \times \text{maturity} \times \text{spawning weight}$
Projection begin year	2025
Projection years	14
Simulations	1000
Alternatives	1, 2, 3, 4, 5, 6, 7

Table 9: SPM projection alternatives included in the generated setup file.

SPM Projection Alternatives

Alternative	Name	Description
1	Maximum permissible ABC	Maximum permissible Tier 3 ABC harvest rate.
2	Author-specified ABC	Author-specified adjustment to the Tier 3 ABC harvest rate; the current multiplier is 1.0.
3	Average recent F	Recent average fishing mortality from the Rceattle-derived input file.
4	Alternative SPR rate	User-specified SPR rate of 0.60.
5	No fishing	Zero fishing mortality after the fixed-catch years.
6	OFL threshold determination	OFL harvest rate for threshold and status-determination calculations.
7	Status-determination ramp	Maximum permissible ABC for three years followed by the OFL status-determination calculation.

6.1 Seven Tier 3 Scenario Results

Catches for 2025 and 2026 are fixed at 1,350 thousand t in every alternative; the alternatives therefore first diverge in 2027 (Table 10). The reported values are simulation means from 1,000 SPM simulations.

Table 10: Tier 3 projection results for all seven FMP alternatives using inputs generated from the Rceattle EBS pollock model. Catch and biomass quantities are in thousand t; $B/B_{35\%}$ is spawning biomass relative to the Tier 3 $B_{35\%}$ proxy.

Seven Tier 3 Projection Alternatives

Alt	Scenario	Year	Catch	ABC	OFL	SSB	F	B/B35
1	Maximum permissible ABC	2027	1,864	1,864	2,259	2,476	0.350	1.17
1	Maximum permissible ABC	2028	1,308	1,308	1,583	2,115	0.310	1.00
2	Author-specified ABC	2027	1,864	1,864	2,259	2,476	0.350	1.17
2	Author-specified ABC	2028	1,308	1,308	1,583	2,115	0.310	1.00
3	Average recent F	2027	1,132	1,132	2,259	2,582	0.200	1.22
3	Average recent F	2028	1,015	1,015	1,963	2,446	0.200	1.16
4	Alternative SPR rate	2027	856	856	2,259	2,619	0.150	1.24
4	Alternative SPR rate	2028	803	803	2,103	2,585	0.150	1.22
5	No fishing	2027	0	0	2,259	2,726	0.000	1.29
5	No fishing	2028	0	0	2,455	3,034	0.000	1.44
6	OFL threshold determination	2027	1,561	1,561	1,561	2,017	0.370	0.95
6	OFL threshold determination	2028	1,210	1,210	1,210	1,831	0.340	0.87
7	Status-determination ramp	2027	1,509	1,809	1,809	2,211	0.320	1.05
7	Status-determination ramp	2028	1,424	1,424	1,424	1,977	0.360	0.94

Table 12: Candidate ABC projection given assumed future catches. Catch is fixed at 1,300 thousand t; ABC is the maximum permissible ABC calculated by SPM. $B/B_{35\%}$ is mean spawning biomass relative to the Tier 3 proxy.

Candidate ABC Projection Given Assumed Future Catches

Year	Catch	ABC	OFL	Mean B	$B/B_{35\%}$
2025	1,300	2,360	2,878	3,523	167%
2026	1,300	2,131	2,588	3,013	143%
2027	1,300	1,891	2,295	2,597	123%
2028	1,300	1,583	1,909	2,366	112%
2029	1,300	1,446	1,763	2,364	112%
2030	1,300	1,459	1,786	2,446	116%
2031	1,300	1,520	1,871	2,547	120%
2032	1,300	1,594	1,967	2,644	125%

Table 11: Age-specific schedules applied in SPM. Natural mortality, maturity, and weights come from the terminal Rceattle configuration; fishery selectivity is the arithmetic mean at age over 2020–2024.

Projection Age-Specific Schedules

Age	Spawning_weight	Fishery_weight	Maturity	Natural_mortality	Selectivity
1	0.085	0.025	0.000	0.900	0.000
2	0.196	0.186	0.008	0.450	0.061
3	0.322	0.401	0.289	0.300	0.220
4	0.534	0.667	0.641	0.300	0.414
5	0.621	0.663	0.842	0.300	0.775
6	0.707	0.720	0.901	0.300	1.031
7	0.822	0.791	0.947	0.300	1.302
8	0.946	0.905	0.963	0.300	1.530
9	1.051	1.009	0.970	0.300	1.456
10	1.113	1.061	1.000	0.300	1.394
11	1.178	1.138	1.000	0.300	1.406
12	1.290	1.257	1.000	0.300	1.353
13	1.332	1.267	1.000	0.300	1.353
14	1.367	1.290	1.000	0.300	1.353
15	1.415	1.368	1.000	0.300	1.353

6.2 Alternative 2 Fixed-Catch Projection

This additional run applies Alternative 2 and fixes catch at 1,300 thousand t for 2025–2032. The projection horizon includes every fixed-catch year.

6.3 Projection Figures

The figures summarize the 1,000 simulations for each alternative. Lines show simulation medians and ribbons show central 90% intervals.

Table 13: Mean SPM results by alternative and projection year. Biomass, catch, ABC, and OFL are in thousand t.

SPM Projection Means

Alt	Year	B	Catch	ABC	OFL	F	$B/B_{35\%}$
1	2025	3,516	1,350	2,360	2,878	0.188	166%
1	2026	2,986	1,350	2,116	2,570	0.210	141%
1	2027	2,476	1,864	1,864	2,259	0.354	117%
1	2028	2,115	1,308	1,308	1,583	0.306	100%
1	2029	2,179	1,265	1,265	1,545	0.303	103%
1	2030	2,321	1,365	1,365	1,671	0.311	110%
1	2031	2,426	1,464	1,464	1,793	0.318	115%
1	2032	2,498	1,531	1,531	1,874	0.322	118%
1	2033	2,532	1,564	1,564	1,913	0.323	120%
1	2034	2,526	1,567	1,567	1,916	0.322	120%
1	2035	2,518	1,564	1,564	1,912	0.323	119%
1	2036	2,504	1,554	1,554	1,900	0.322	118%
1	2037	2,498	1,549	1,549	1,893	0.322	118%
1	2038	2,511	1,551	1,551	1,896	0.321	119%
2	2025	3,516	1,350	2,360	2,878	0.188	166%
2	2026	2,986	1,350	2,116	2,570	0.210	141%
2	2027	2,476	1,864	1,864	2,259	0.354	117%
2	2028	2,115	1,308	1,308	1,583	0.306	100%
2	2029	2,179	1,265	1,265	1,545	0.303	103%
2	2030	2,321	1,365	1,365	1,671	0.311	110%
2	2031	2,426	1,464	1,464	1,793	0.318	115%
2	2032	2,498	1,531	1,531	1,874	0.322	118%
2	2033	2,532	1,564	1,564	1,913	0.323	120%
2	2034	2,526	1,567	1,567	1,916	0.322	120%
2	2035	2,518	1,564	1,564	1,912	0.323	119%
2	2036	2,504	1,554	1,554	1,900	0.322	118%
2	2037	2,498	1,549	1,549	1,893	0.322	118%
2	2038	2,511	1,551	1,551	1,896	0.321	119%
3	2025	3,516	1,350	1,420	2,878	0.188	166%
3	2026	2,986	1,350	1,283	2,570	0.210	141%
3	2027	2,582	1,132	1,132	2,259	0.198	122%
3	2028	2,446	1,015	1,015	1,963	0.198	116%
3	2029	2,569	1,001	1,001	1,950	0.198	122%
3	2030	2,754	1,057	1,057	2,085	0.198	130%
3	2031	2,915	1,124	1,124	2,239	0.198	138%
3	2032	3,045	1,182	1,182	2,364	0.198	144%
3	2033	3,131	1,227	1,227	2,457	0.198	148%
3	2034	3,164	1,249	1,249	2,503	0.198	150%
3	2035	3,179	1,258	1,258	2,523	0.198	150%
3	2036	3,179	1,261	1,261	2,528	0.198	150%
3	2037	3,178	1,261	1,261	2,529	0.198	150%
3	2038	3,193	1,263	1,263	2,533	0.198	151%
4	2025	3,516	1,350	1,071	2,878	0.188	166%
4	2026	2,986	1,350	970	2,570	0.210	141%
4	2027	2,619	856	856	2,259	0.146	124%
4	2028	2,585	803	803	2,103	0.146	122%

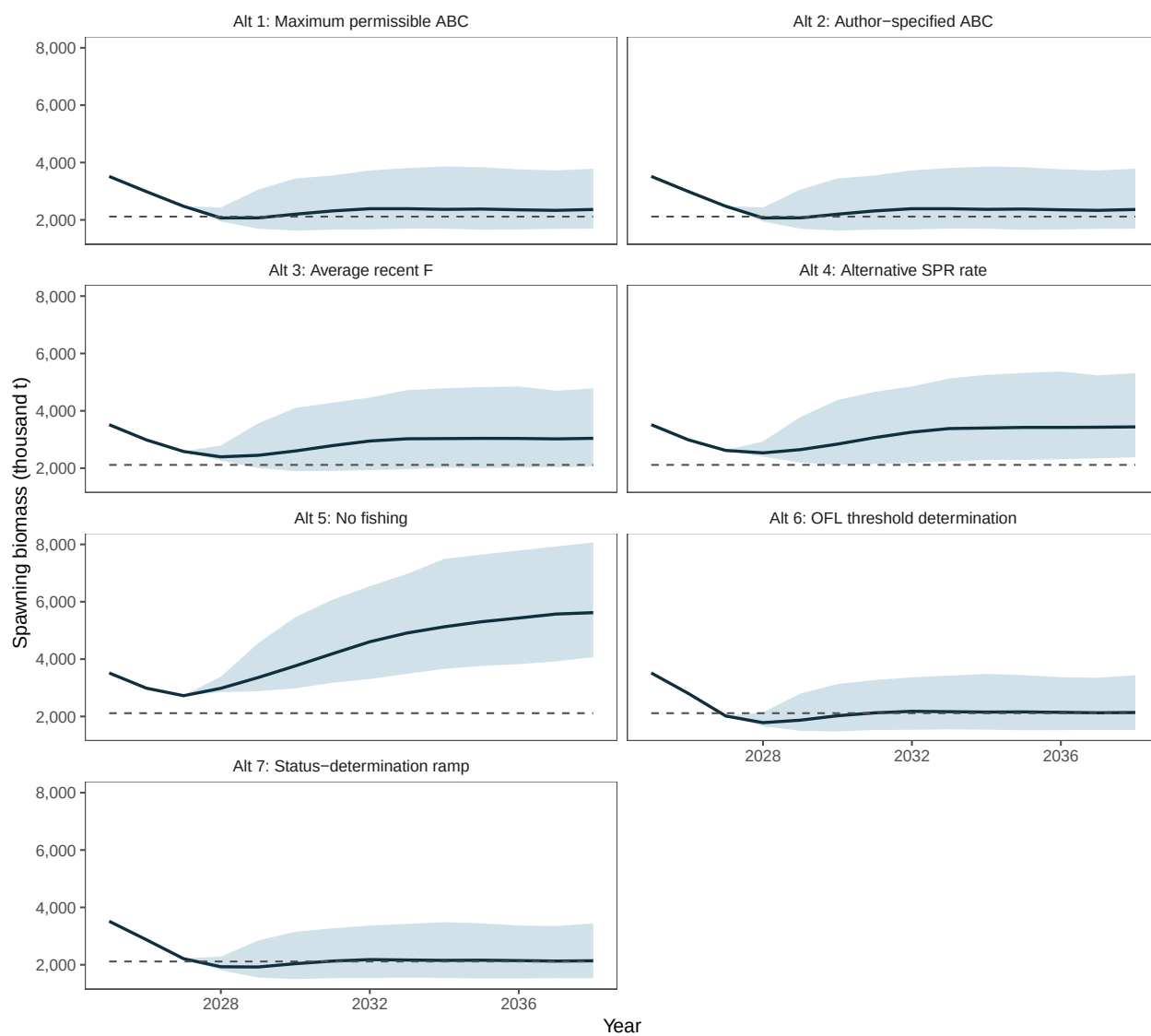


Figure 6: Projected spawning biomass under all seven Tier 3 alternatives. Lines are simulation medians, ribbons span the 5th to 95th percentiles, and dashed lines mark B35%.

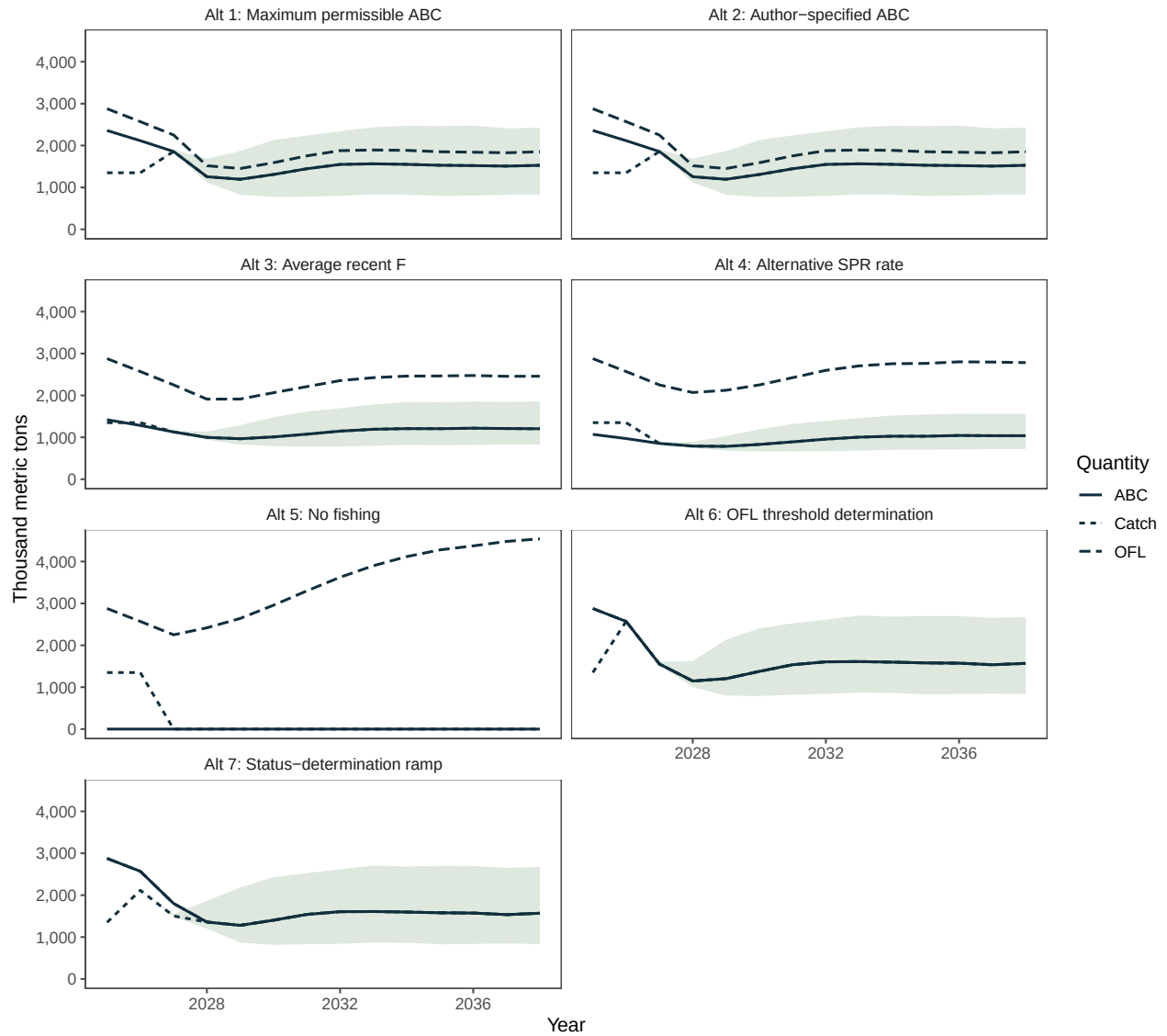


Figure 7: Projected catch, ABC, and OFL under all seven Tier 3 alternatives. Lines are simulation medians and catch ribbons span the 5th to 95th percentiles.

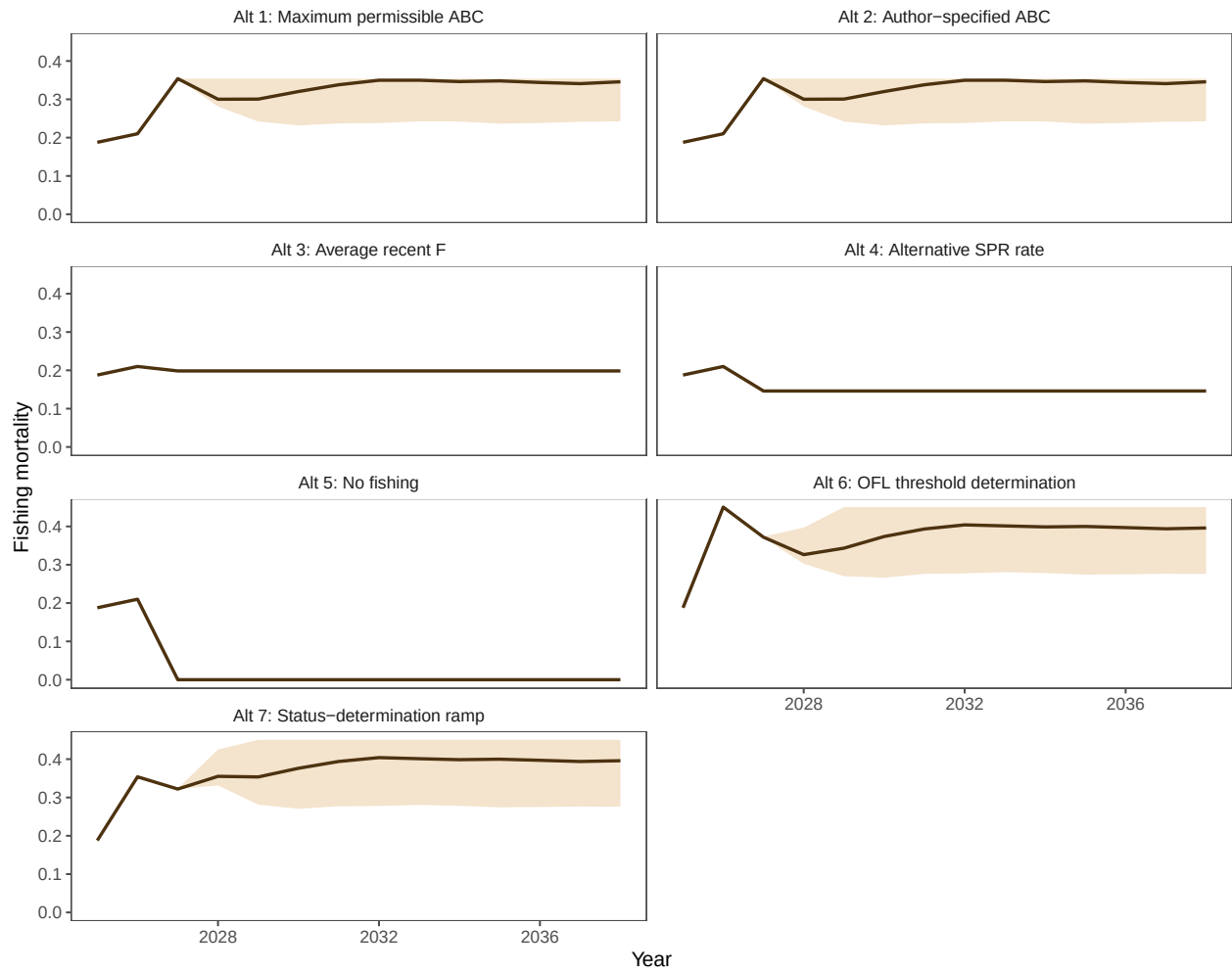


Figure 8: Projected fishing mortality under all seven Tier 3 alternatives. Lines are simulation medians and ribbons span the 5th to 95th percentiles.

7 Exploratory DSEM Recruitment Pathways

This section is organized around causal pathways, the covariates used to represent them, and the comparison approach. It adapts the bottom-up energy, top-down overlap, transport, and switching-control hypotheses developed in the [EBS pollock Early Survival Pathways synthesis](#) ([J. N. Ianelli 2026](#)). The Rceattle experiments target the age-1 recruitment state, whereas the broader synthesis follows cohorts through survival to age 3. Some annual sea-ice, spring-SST, and juvenile-predator-overlap series used below were provided by the EcoState case-study work ([Goodman 2026](#)). EcoState supplied covariate products; the Rceattle fits and their interpretation were developed separately for this working paper.

7.1 Causal graph proof of concept and covariate roles

The integration of DSEM within the Rceattle package is an enhancement that we think provides greater facility to account for broader ecological aspects within a management framework. Environmentally informed recruitment models benefit from covariates linked to plausible mechanisms, evaluated at relevant temporal and spatial scales, and tested for predictive and management value; otherwise added complexity can produce unstable or spurious relationships ([Haltuch et al. 2019](#)). DSEM provides a computational framework for representing simultaneous and lagged ecological pathways in incomplete multivariate time series ([Thorson et al. 2024](#)). Here, causal-method testing means comparing prespecified graphs rather than searching for the strongest individual correlation. The tests ask whether the data are consistent with the graph’s implied conditional-independence relationships, whether added edges improve prediction, and whether conclusions persist across alternative cohort timing, proxy, and missing-data assumptions ([Thorson et al. 2025](#)). A recent Gulf of Alaska pollock application embedded DSEM within a state-space stock assessment and found that an intermediate-complexity causal model reduced unexplained recruitment variance and improved one-year-ahead forecasts, illustrating the value of parsimonious, hypothesis-driven model comparison ([Champagnat et al. 2026](#)).

The conceptual pathway links spawning and environmental conditions to recruitment while identifying the two covariates used here within that broader context (Figure 9). The current proof of concept estimates only the two dashed paths to age-1 recruitment; the solid paths identify mechanisms and covariates for future expansion rather than fitted paths in this analysis.

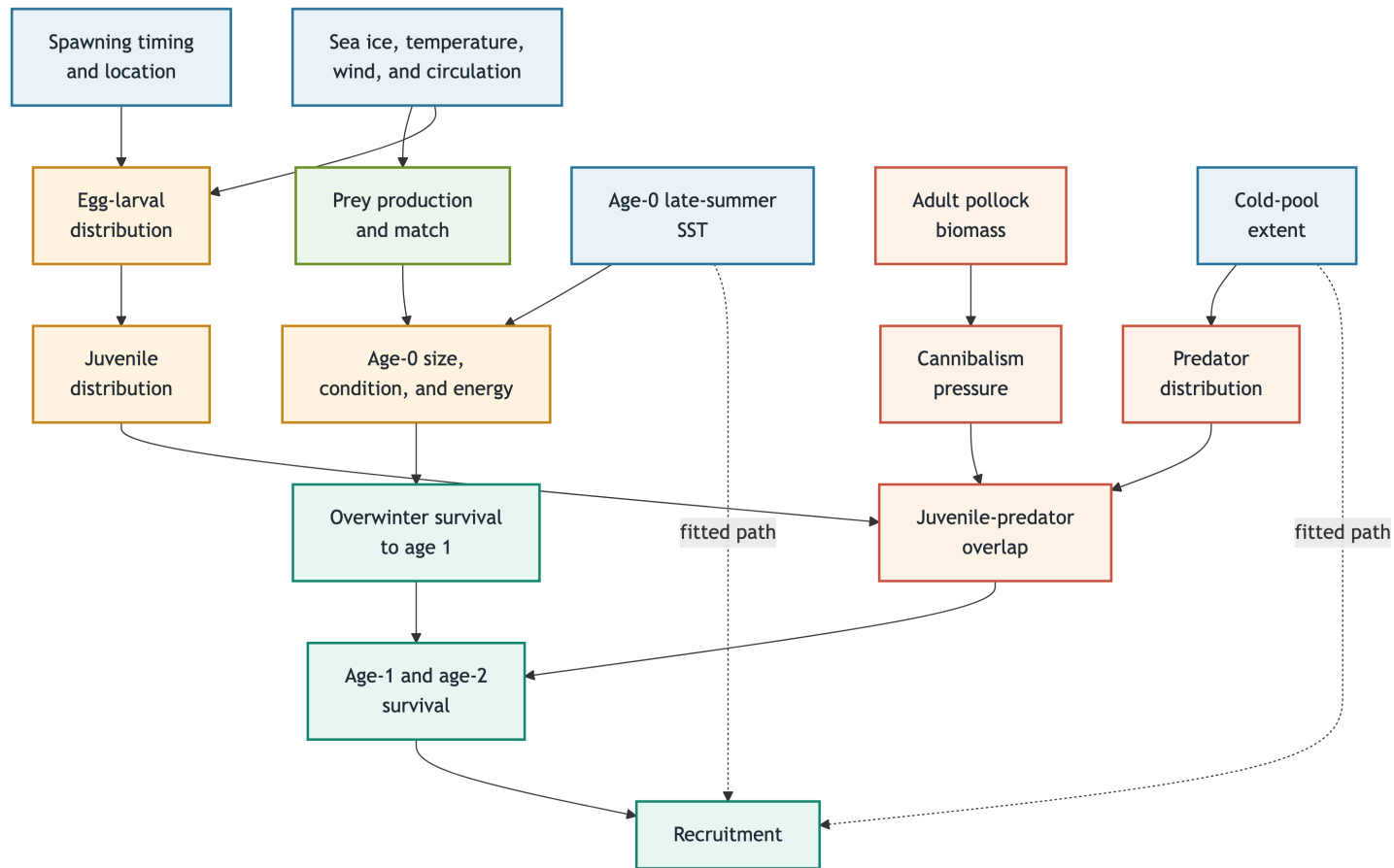


Figure 9: Compact conceptual directed acyclic graph for eastern Bering Sea pollock early survival. Solid arrows summarize broader mechanisms that motivate future covariates. Dashed arrows identify the two direct environmental effects estimated in the current DSEM proof of concept: age-0 late-summer SST and age-1-year cold-pool extent on age-1 recruitment.

Table 14: Causal pathways, their representation in the exploratory Rceattle-DSEM runs, and the next measurements needed to represent each pathway more directly.

Pathway	Representation in current runs	Next direct measurements
Bottom-up energy	Cold-pool extent; late-summer SST; maximum winter sea ice; April–May SST	Sea-ice days after February; lipid-rich copepods; fall age-0 energy; bloom or prey-match timing
Top-down overlap	Arrowtooth and adult-pollock range-overlap indices	Uncertainty-aware overlap; predation mortality; age-specific juvenile distributions
Transport and match	Physical covariates serve only as indirect proxies	Wind and circulation; spawning distribution; larval-delivery and spatial prey-match indices
Switching control	Separate physical and overlap suites with stationary coefficients	Interactions or regime models; rolling stability; blocked prediction

The dashed arrows in Figure 9 identify the direct environmental paths in the initial proof of concept. The solid paths organize covariates by mechanism and show why a temperature or sea-ice index is an upstream proxy rather than a measured energy, transport, or predation process. The two later suites test alternative physical proxies and observed range-overlap indices; they do not fit the full mediated graph in Figure 9.

7.2 Cohort alignment and model-comparison approach

The proof of concept treats recruitment deviations as random effects and aligns covariates with the year in which each cohort enters the model at age 1. The spawning-biomass proxy was removed. The two standardized environmental series are cold-pool extent in the age-1 calendar year (42 observations) and late-summer sea surface temperature (SST) in the cohort’s age-0 year (41 observations). The age-0 SST value is carried forward to the DSEM row for that cohort’s age-1 recruitment; the zero-lag path is therefore cohort-aligned rather than a claim that SST was measured in the age-1 year.

In this section “IID” refers to the no-recruitment-edge reference model.

The comparisons can therefore be read as three increasingly specific questions:

- **IID reference:** how much recruitment variation remains when neither candidate is connected directly to recruitment?
- **One-edge model:** does adding one specified arrow explain enough of the between-year recruitment pattern to improve the model?
- **Two-edge model:** does the second arrow add information after accounting for the first, or are the two covariates largely describing the same years?

Each arrow is a statistical test of the biological hypothesis represented by that covariate, not proof of the full mechanism in the causal graph. Models within a suite use the same observation years, missingness pattern, standardization, and AR1 covariate processes. AIC is therefore interpreted only within a suite. The tables also report the percentage by which each model’s estimated residual recruitment variance is lower than its suite’s IID reference. A positive value means less recruitment variation remains unexplained; zero identifies the IID baseline. This is an in-sample variance comparison, not a causal R-squared or a predictive-skill score. Path intervals, convergence diagnostics, randomized controls, and blocked or terminal-year prediction remain necessary before a pathway is considered supported.

7.3 Introducing alternative covariates

In plain language, these four models ask whether recruitment is better described with no environmental edge, a cold-pool edge, an age-0 late-summer SST edge, or both edges estimated together.

Table 15: Exploratory DSEM recruitment-covariate models based on Rceattle 5.8.1. Delta AIC and the percentage reduction in residual recruitment variance are calculated relative to models using the same years and covariate observations.

Model	AIC	Delta AIC	Residual variation reduced (%)
Age-0 late-summer SST	4,144.56	0.00	10.5
Cold pool + age-0 late-summer SST	4,146.19	1.63	12.0
IID recruitment deviations	4,146.81	2.25	0.0
Cold pool	4,148.66	4.11	0.3

All four initial fits reached OK status with positive-definite Hessians and maximum absolute gradients no larger than $9.4\text{e-}05$. The model containing only age-0 late-summer SST had the lowest AIC (Table 15; Figure 10). Its estimated residual recruitment variance was 10.5% lower than in the IID model; adding cold pool changed that reduction only to 12.0%. The cold-pool recruitment coefficient was small and its interval included zero in both the single-covariate and combined models. The SST coefficient was negative and its interval excluded zero in both configurations. This is a modest in-sample signal: the SST-only model improved AIC by 2.25 units relative to IID, but the result has not yet passed a blocked or terminal-year predictive test.

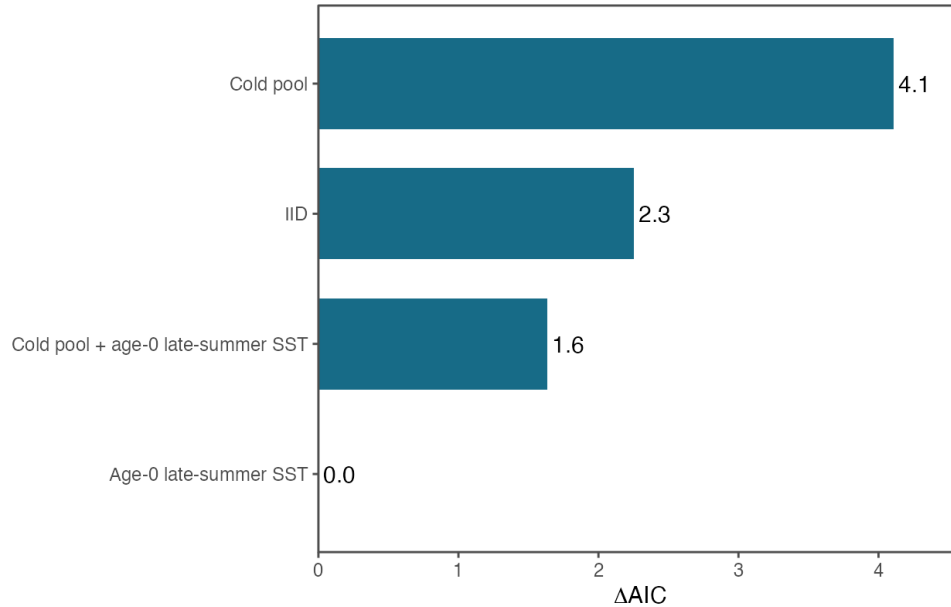


Figure 10: Akaike information criterion differences for the four exploratory DSEM recruitment models. The age-0 late-summer SST model has the lowest AIC; bars show the increase in AIC relative to that model.

Table 16: Standardized DSEM paths to age-1 recruitment. Intervals are estimate plus or minus 1.96 standard errors.

Covariate	Estimate	Standard error	Lower 95%	Upper 95%
Cold pool				
Cold pool	0.041	0.107	-0.169	0.251
Age-0 late-summer SST				
Age-0 late-summer SST	-0.222	0.104	-0.426	-0.017
Cold pool + age-0 late-summer SST				
Cold pool	-0.069	0.114	-0.292	0.153
Age-0 late-summer SST	-0.254	0.118	-0.485	-0.024

The age-0 late-summer SST effect is negative and separated from zero, whereas the cold-pool effect remains near zero (Figure 11). This direction is consistent with the hypothesized warm-year energetic constraint on age-0 pollock, but the fitted path does not by itself identify the intervening prey, condition, or overwinter-survival mechanism (Heintz et al. 2013; Duffy-Anderson et al. 2016). The resulting recruitment trajectories remain visually similar despite the AIC differences (Figure 12).

As a randomization check, the analysis replaced both observed environmental series with independent standard-normal values while preserving their missing-year patterns (seed 20260812). None of the randomized recruitment paths improved on the randomized IID model: their delta AIC values ranged from 1.03 to 2.95, and every randomized path interval included zero. This result shows that the same procedure did not identify an effect in these randomized series, but it does not replace out-of-sample validation.

7.4 Covariate suites

The experiments use three small, biologically prespecified covariate suites. The initial environmental suite compares cold-pool extent with age-0 late-summer SST. The physical-forcing suite compares maximum winter regional sea ice with April–May southeastern Bering Sea SST. The top-down suite compares juvenile-pollock range overlap with arrowtooth flounder and adult pollock. The latter four annual series were supplied by the EcoState case-study repository (Goodman 2026); they serve only as covariate inputs to the separate Rceattle-DSEM runs reported here.

The overlap products are occurrence-range indices rather than biomass-weighted overlap or direct predation mortality. The adult-pollock index is also partly population-derived. These properties, and the missing uncertainty across the 100 occurrence simulations used to construct the means, limit causal interpretation. The physical and overlap suites are therefore tests of direct proxy paths in the larger causal graph rather than estimates of the intervening energy or mortality processes.

7.5 Physical-forcing pathway

The physical-candidate suite is a separate Rceattle-DSEM run from the earlier cold-pool and late-summer-SST analysis. It compares IID recruitment deviations, maximum winter regional sea-ice extent, April–May southeastern Bering Sea SST, and both physical candidates. The source-year values describe the cohort’s age-0 winter and spring and are carried to the following age-1 DSEM row. The model contains 45 finite sea-ice values and 41 finite spring-SST values; the 41 paired values have correlation -0.699. AIC comparisons are valid within this suite; they should not be compared directly with AIC from a run containing a different environmental observation block.

In plain language, the comparison asks whether recruitment is better described with no physical edge, a winter-sea-ice edge, a spring-SST edge, or both edges. The combined model asks whether each physical

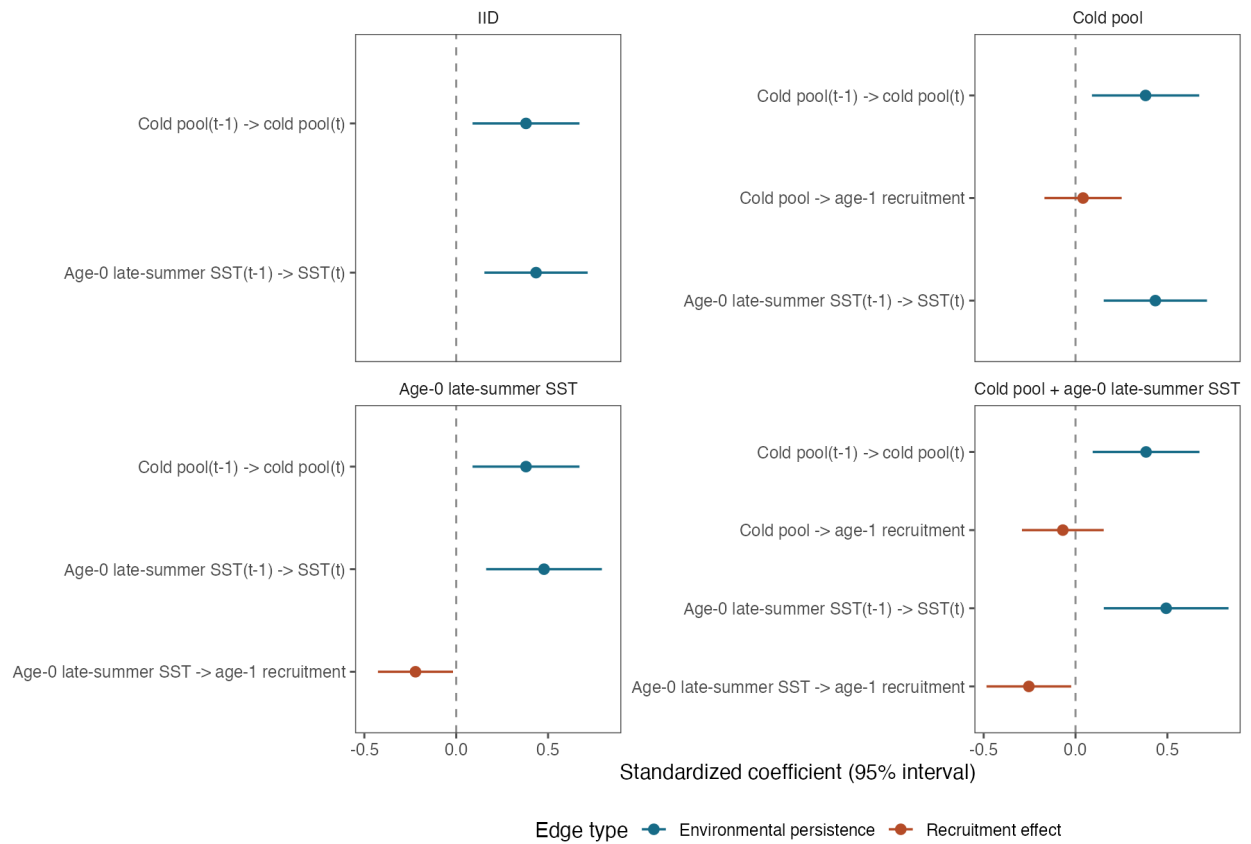


Figure 11: Standardized directed-edge coefficients for the four exploratory DSEM models. Points are estimates and horizontal lines are 95% intervals. Blue edges describe temporal persistence in the observed covariates; orange edges describe direct effects on age-1 recruitment.

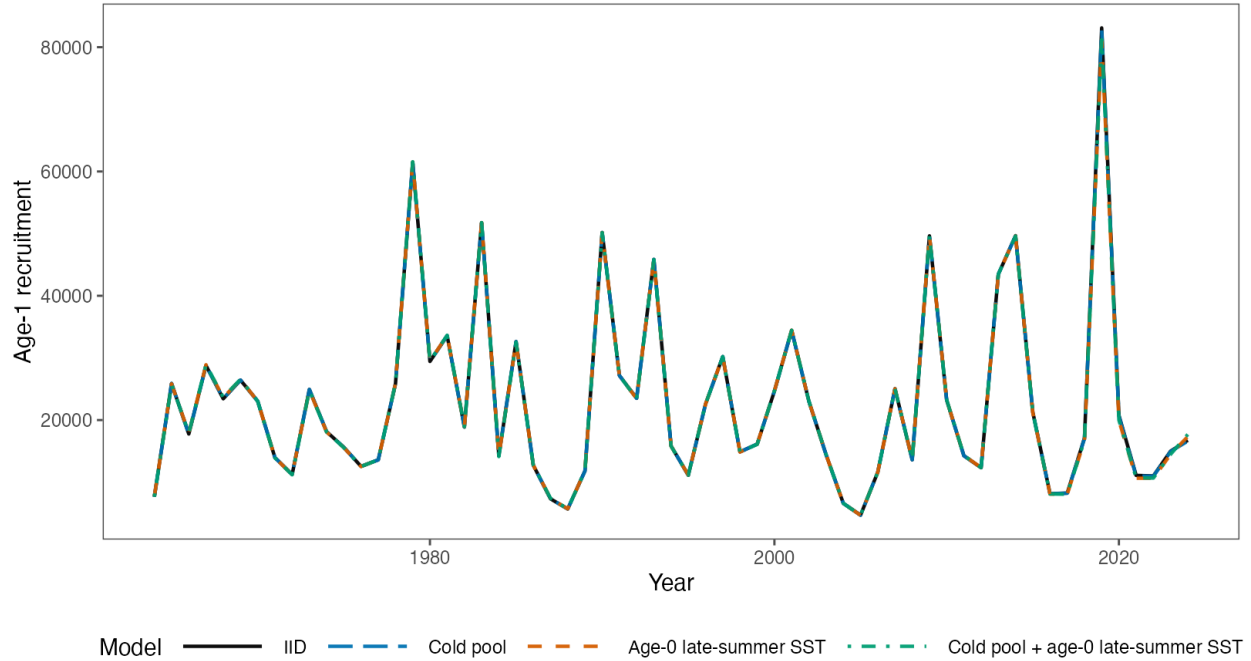


Figure 12: Estimated age-1 recruitment from the IID, cold-pool, age-0 late-summer SST, and combined exploratory DSEM models.

proxy contributes after the other is already included; it does not assume that both arrows represent separate mechanisms.

The model graph includes AR(1) persistence for each observed covariate and tests either or both dashed paths to age-1 recruitment (Figure 13). The sea-ice-to-SST arrow is a mechanistic hypothesis shown for interpretation; the fitted combined model estimates two direct recruitment paths rather than that intermediate edge.

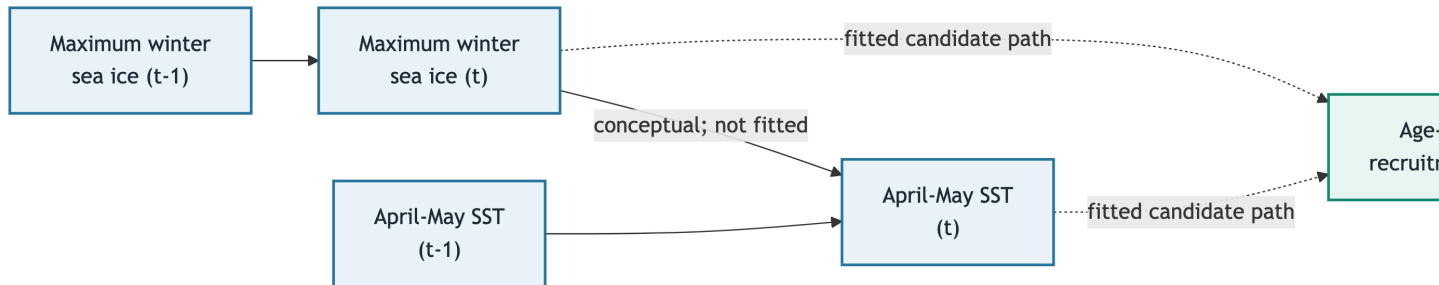
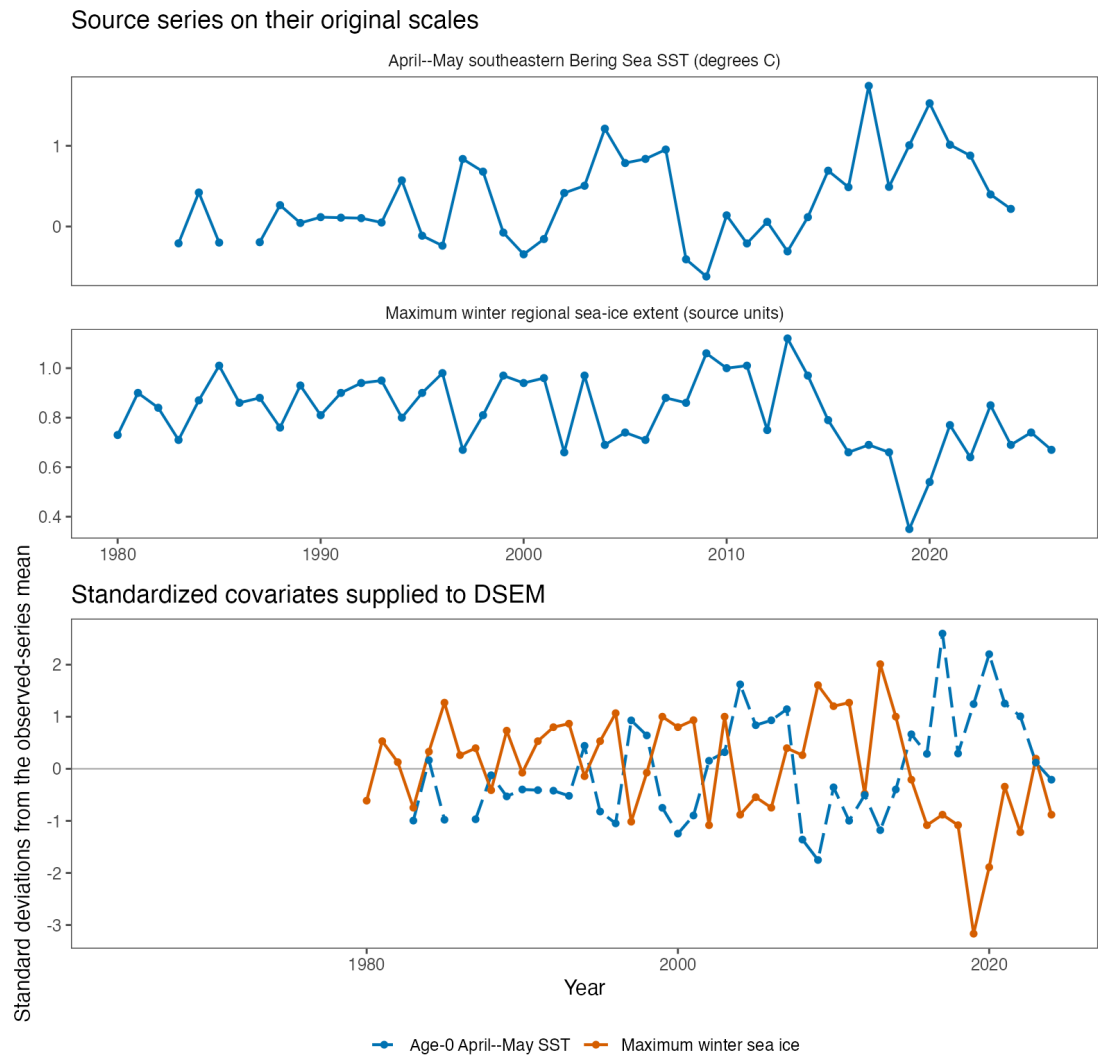


Figure 13: Directed graph for the physical-forcing Rceattle-DSEM suite. Solid horizontal arrows are fitted AR(1) covariate-persistence paths. Dashed arrows are the direct recruitment paths fitted singly and jointly. The sea-ice-to-SST arrow is a conceptual mechanism and is not estimated in this run.



Maximum winter regional sea-ice extent observations: 45; Age-0 April–May southeastern Bering Sea SST observations: 41.

Figure 14: Physical-forcing source series and the standardized, cohort-aligned values supplied to the Rceattle-DSEM suite.

Table 17: Separate Rceattle-DSEM physical-candidate model comparison. Delta AIC and the percentage reduction in residual recruitment variance are calculated only within the common physical-candidate observation block.

Model	AIC	Delta AIC	Residual variation reduced (%)
Both physical candidates	4,134.32	0.00	27.9
Age-0 April–May SST	4,149.85	15.53	10.5
IID recruitment deviations	4,152.08	17.76	0.0
Maximum winter sea ice	4,153.20	18.88	1.9

All four physical-candidate fits reached OK status with positive-definite Hessians and maximum absolute gradients no larger than $1.3\text{e-}04$ (Table 17). The combined model had the lowest AIC and reduced estimated unexplained recruitment variance by 27.9% relative to IID. April–May SST was negative in its single-path fit and in the combined fit. Maximum winter sea ice was weakly positive by itself, then negative with an interval spanning zero in the combined fit. This sign change, together with the strong inverse correlation between the two covariates, indicates a suppression pattern; the joint AIC improvement should not be interpreted as two independently identified causal effects.

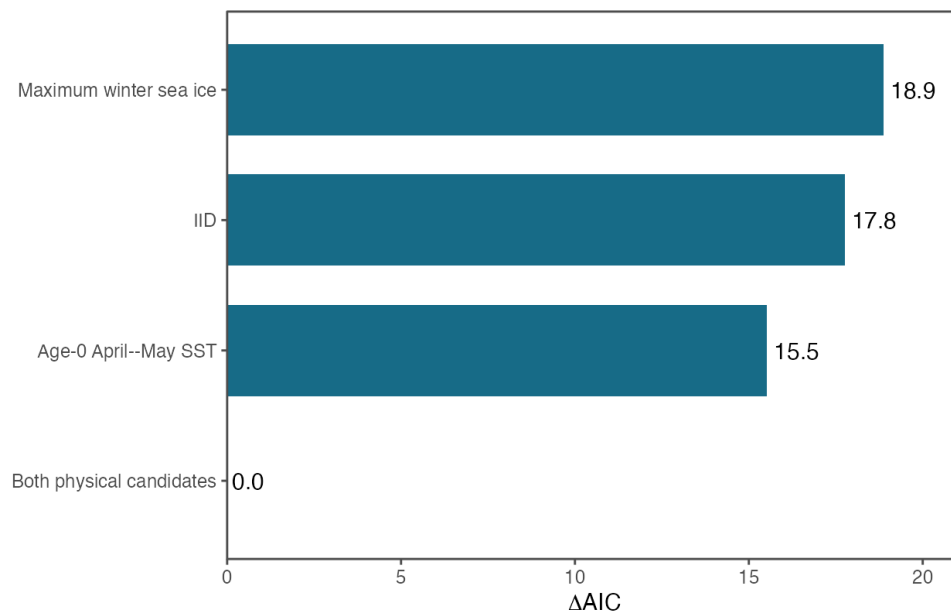


Figure 15: Akaike information criterion differences for the four models in the physical-forcing Rceattle-DSEM suite.

Table 18: Standardized direct paths to age-1 recruitment in the physical-forcing Rceattle-DSEM suite.

Covariate	Estimate	Standard error	Lower 95%	Upper 95%
Maximum winter sea ice				
Maximum winter sea ice	0.099	0.105	-0.107	0.305
Age-0 April–May SST				
Age-0 April–May SST	-0.230	0.109	-0.444	-0.016
Both physical candidates				
Maximum winter sea ice	-0.131	0.101	-0.330	0.067
Age-0 April–May SST	-0.359	0.085	-0.525	-0.193

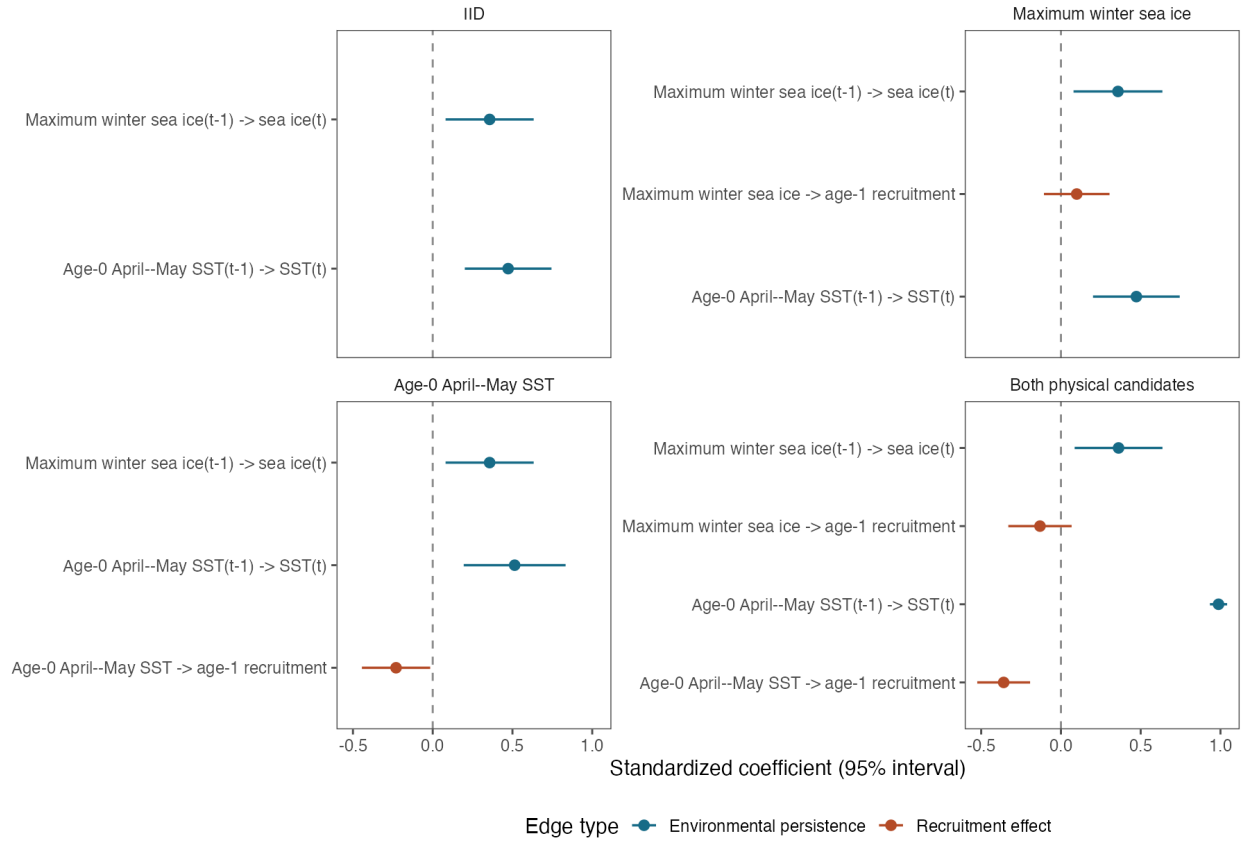


Figure 16: Standardized directed-edge coefficients for the physical-forcing Rceattle-DSEM suite. Points are estimates and horizontal lines are 95% intervals.

The matching missingness-preserving randomization control used independent standard-normal values with seed 20260812. Every randomized physical-candidate model had higher AIC than its IID model; the AIC increases ranged from 1.48 to 3.43 units, and every randomized recruitment-path interval included zero.

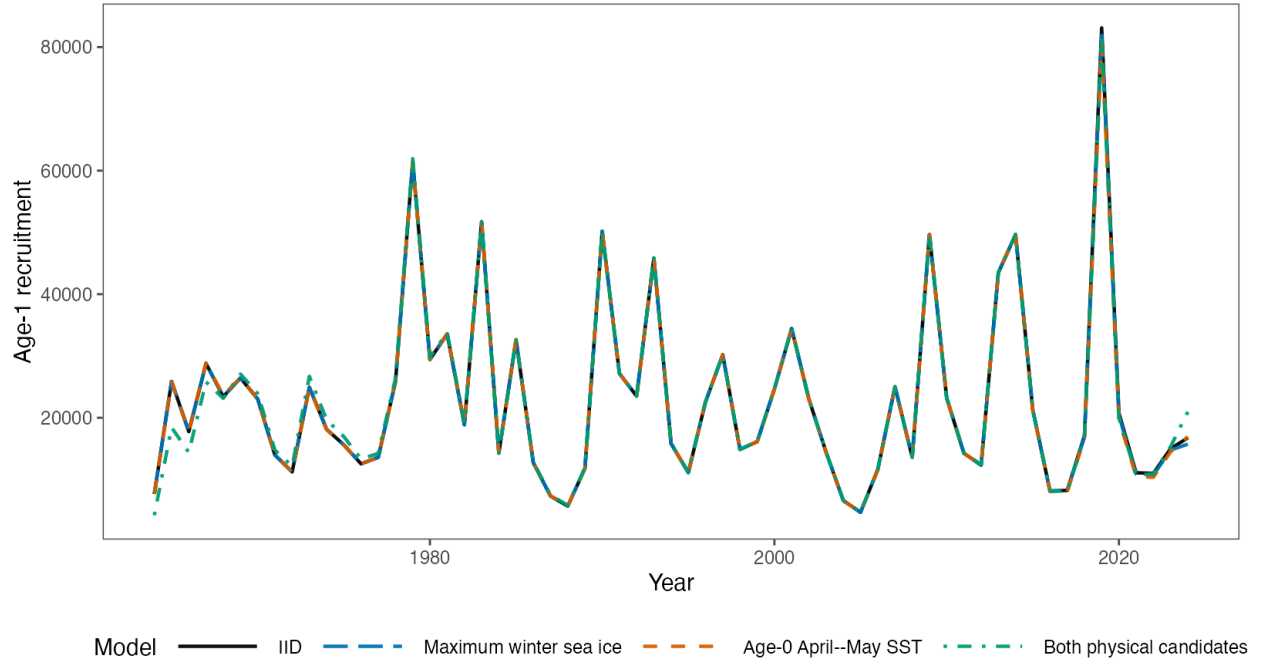


Figure 17: Estimated age-1 recruitment from the four models in the physical-forcing Rceattle-DSEM suite.

7.6 Top-down overlap pathway

The second separate run compares IID recruitment deviations with annual arrowtooth–juvenile pollock range overlap, adult-pollock–juvenile pollock range overlap, and both overlap indices. Each index has 40 observations from 1982–2022 with the 2020 survey gap, and their paired correlation is 0.480. The indices are aligned to the same calendar year as the age-1 recruitment state. Because the bottom-trawl survey occurs after the start-of-year state, these are contemporaneous exploratory associations rather than established predation effects.

In plain language, the four models ask whether recruitment is better described with no overlap edge, an arrowtooth-overlap edge, an adult-pollock-overlap edge, or both edges. The combined model asks whether arrowtooth overlap adds information after adult-pollock overlap is known, and vice versa; it does not estimate predation or cannibalism mortality directly.

The graph distinguishes the direct paths actually fitted from the intervening predation and cannibalism mechanisms (Figure 18). The run does not estimate predation mortality, and the adult-pollock overlap index remains partly population-derived.

Table 19: Separate Rceattle-DSEM juvenile-predator-overlap model comparison. Delta AIC and the percentage reduction in residual recruitment variance are calculated only within the common overlap observation block.

Model	AIC	Delta AIC	Residual variation reduced (%)
Adult-pollock overlap	4,076.37	0.00	20.8
Both overlap candidates	4,078.05	1.68	20.9
Arrowtooth overlap	4,080.14	3.77	16.1
IID recruitment deviations	4,096.14	19.77	0.0

All four overlap fits reached OK status with positive-definite Hessians and maximum absolute gradients no

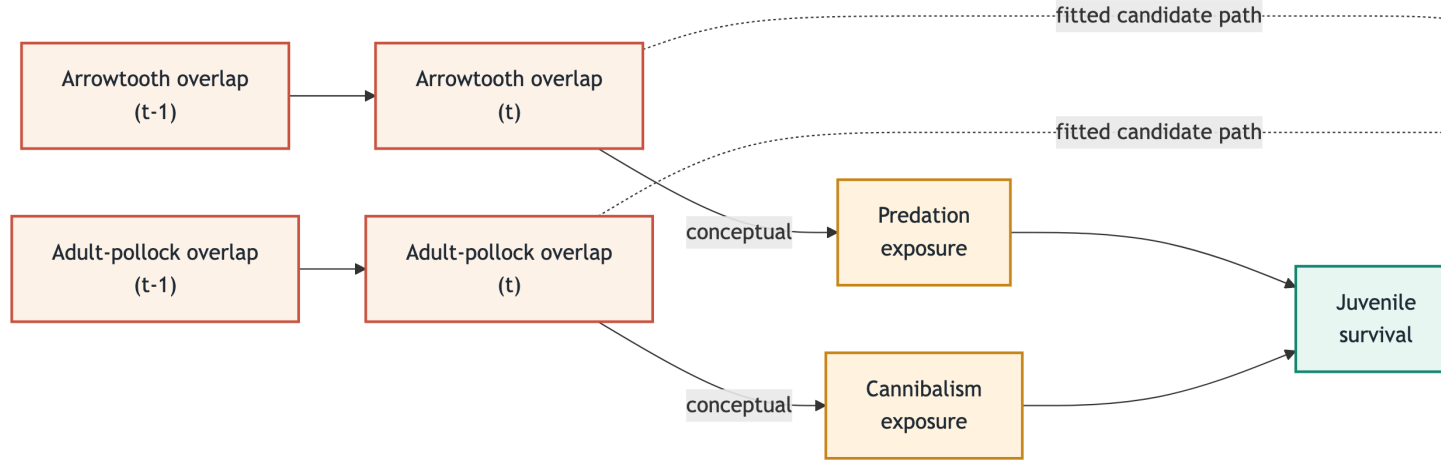


Figure 18: Directed graph for the top-down juvenile-predator-overlap Rceattle-DSEM suite. Solid horizontal arrows are fitted AR(1) persistence paths. Dashed arrows are the direct recruitment paths fitted singly and jointly. Predation and cannibalism nodes show conceptual mechanisms that are not estimated in this run.

larger than $8.8\text{e-}05$ (Table 19). The adult-pollock-overlap model had the lowest AIC and reduced estimated unexplained recruitment variance by 20.8% relative to IID. Each overlap coefficient was negative and separated from zero when fitted alone. In the combined model, the adult-pollock-overlap coefficient remained negative, whereas the arrowtooth-overlap coefficient shifted near zero and its interval included zero. Adding arrowtooth overlap to adult-pollock overlap increased AIC by 1.68 units and produced essentially no further reduction in unexplained recruitment variance.

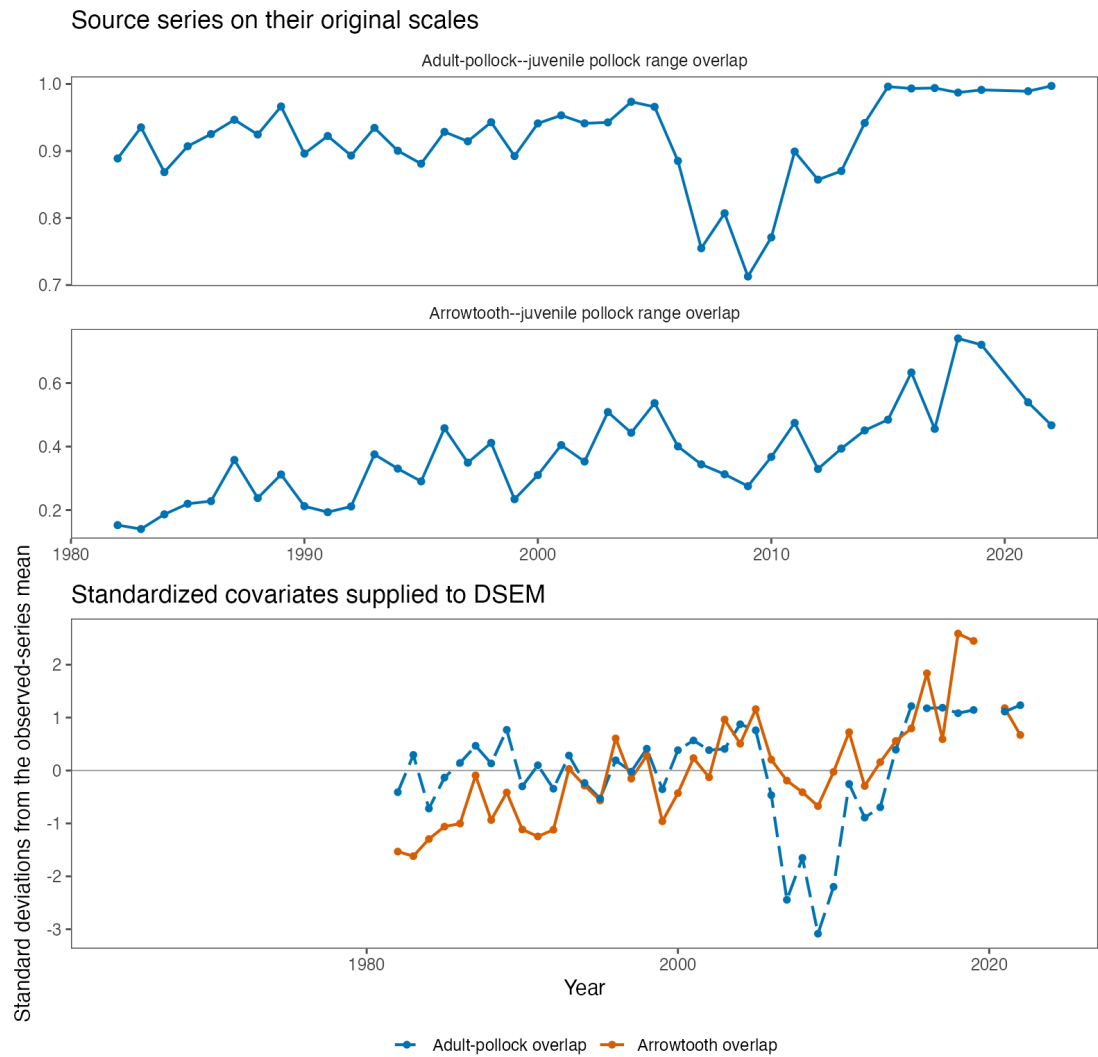
Table 20: Standardized direct paths to age-1 recruitment in the top-down juvenile-predator-overlap Rceattle-DSEM suite.

Covariate	Estimate	Standard error	Lower 95%	Upper 95%
Arrowtooth overlap				
Arrowtooth–juvenile pollock overlap	-0.282	0.067	-0.414	-0.150
Adult-pollock overlap				
Adult-pollock–juvenile pollock overlap	-0.320	0.071	-0.460	-0.181
Both overlap candidates				
Arrowtooth–juvenile pollock overlap	0.061	0.106	-0.147	0.268
Adult-pollock–juvenile pollock overlap	-0.337	0.078	-0.490	-0.184

The overlap randomization control used the same seed and preserved the common 2020 gap. Its random candidate-2 model improved AIC by only 0.93 units relative to randomized IID, and every randomized recruitment-path interval included zero. This small chance improvement is much smaller than the observed adult-pollock-overlap AIC improvement, but it reinforces the need for blocked prediction and timing sensitivities before interpretation.

7.7 Next-stage pathway tests

The next covariates should fill mechanisms in the causal graph rather than add more interchangeable physical proxies. For the bottom-up pathway, sea-ice days after February remains the highest-priority physical



Arrowtooth--juvenile pollock range overlap observations: 40; Adult-pollock--juvenile pollock range overlap observations: 40.

Figure 19: Juvenile-predator range-overlap series and the standardized values supplied to the top-down Rceattle-DSEM suite.

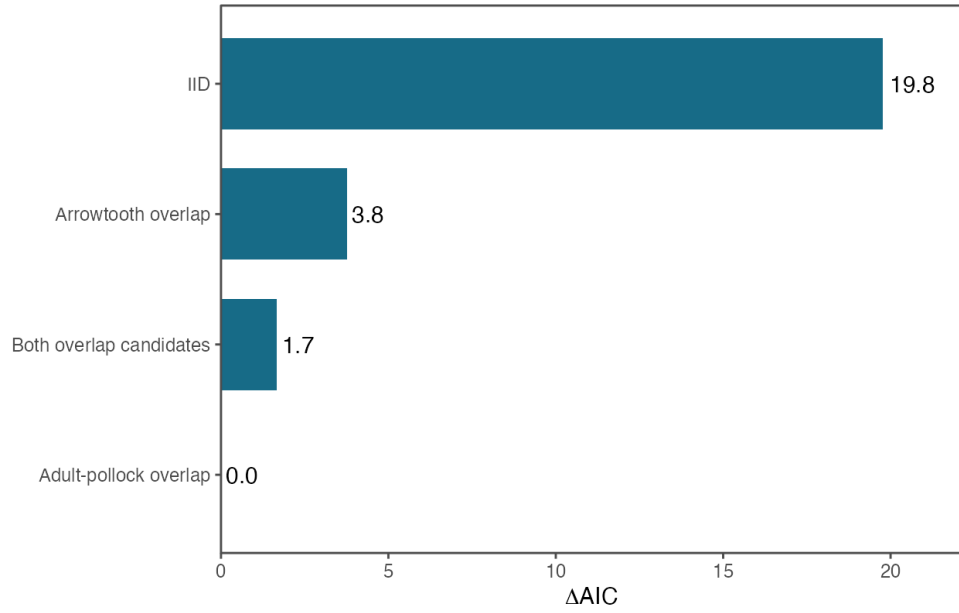


Figure 20: Akaike information criterion differences for the four models in the top-down juvenile-predator-overlap Rceattle-DSEM suite.

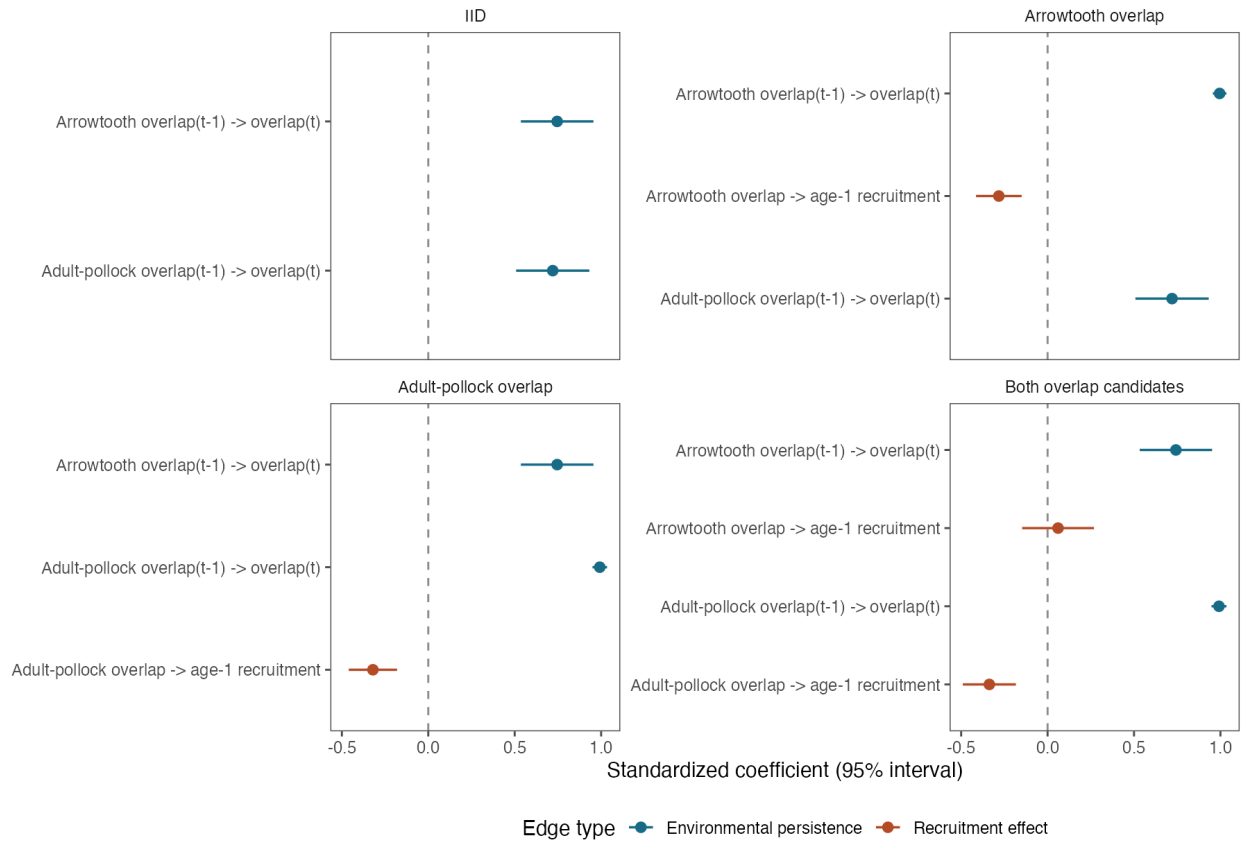


Figure 21: Standardized directed-edge coefficients for the top-down juvenile-predator-overlap Rceattle-DSEM suite. Points are estimates and horizontal lines are 95% intervals.

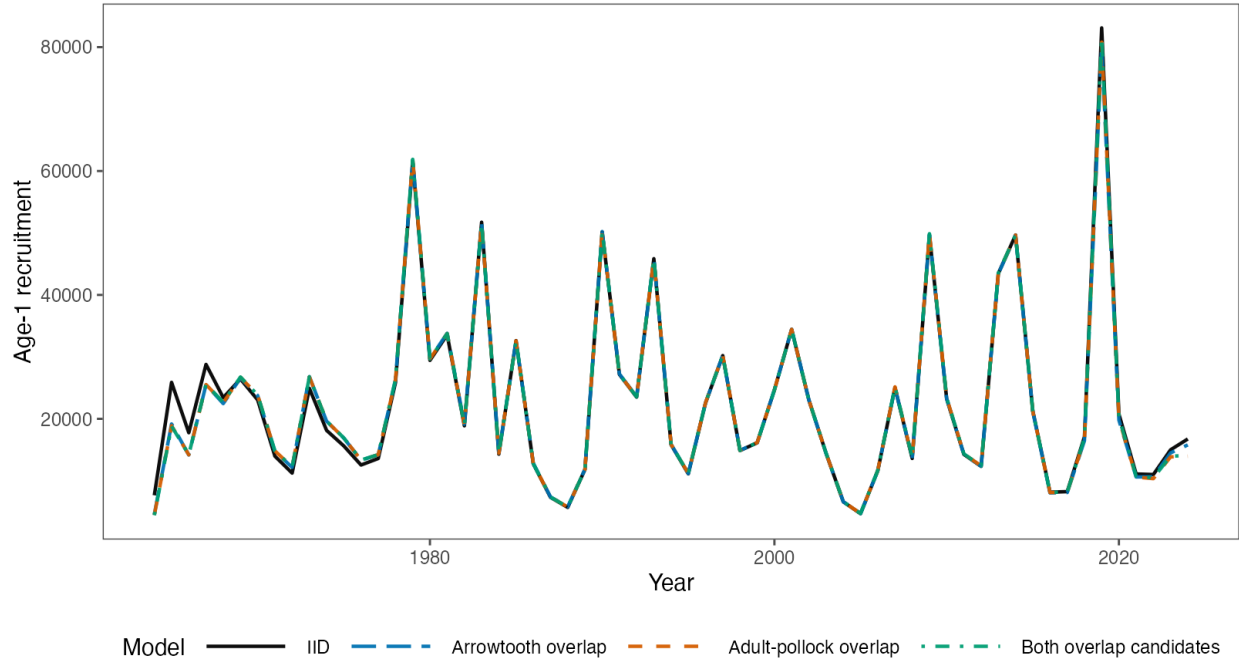


Figure 22: Estimated age-1 recruitment from the four models in the top-down juvenile-predator-overlap Rceattle-DSEM suite.

candidate because it was the best-performing sea-ice metric in a regional analysis, although its relationship with year-class strength differed between historical periods (Hunt et al. 2020). Large lipid-rich copepods provide a more proximal food-web measure, but the current cohort table contains only 15 observations (Eisner et al. 2020). A consistent fall age-0 total-energy or energy-density series would measure the overwinter-survival mediator most directly (Heintz et al. 2013; Siddon et al. 2013). Wind mixing, stratification, bloom timing, and a spatial prey-match index belong upstream of prey and age-0 condition rather than as unstructured additions to recruitment (Gann et al. 2016).

For the top-down pathway, the priority is to propagate uncertainty in the range-overlap indices and develop predator-specific mortality or age-specific juvenile-distribution products. For the transport pathway, the missing measurements are spawning distribution, wind and circulation, and a cohort-specific larval-delivery index. Broad atmospheric indices such as the North Pacific Index or Aleutian Low are most useful as switching-control or regime descriptors after the shelf-scale mechanisms are represented.

The DSEM script is structured so additional cohort-level covariates can be added without altering the assessment configuration. Each expansion should:

- state the recruitment mechanism and cohort timing for each candidate before fitting it;
- preserve a common observation-year universe and document original units, standardization, gaps, and data sources;
- screen correlated covariates, then compare IID, single-path, and limited joint-path models rather than searching many combinations;
- repeat missingness-preserving randomized controls and examine sensitivity to observation error, process assumptions, and priors; and
- use blocked or terminal-year holdouts to test prediction before interpreting an in-sample AIC improvement.

This approach can accommodate additional ESP covariates, but the candidate set should remain biologically prespecified and small relative to the available cohort record. DSEM results should remain research products until the paths are stable across timing assumptions, retrospective peels, and predictive checks.

8 Summary

When Rceattle parameters and fishery selectivity are set to the values used in the ADMB model, Rceattle reproduces spawning biomass and predicted catch to numerical precision. This establishes agreement in the tested population-dynamics calculations and parameter mapping. When Rceattle is fit to the data, the trajectories differ more, and the residual least-identified direction remains in the shared ATS/AVO annual selectivity deviations. The added ATS/AVO coefficient-increment penalty reduced the condition number below the convergence-warning threshold.

The diagnostic results identify a focused development path for the September 2026 Plan Team evaluation. The Rceattle fit reaches a small gradient, an OK convergence status, and a positive-definite Hessian. Composition OSA residuals remain underdispersed. Nine terminal-year-consistent peels converge successfully and produce positive retrospective bias in spawning biomass, total biomass, and recruitment, together with negative bias in fishing mortality. The historical 2D AR1 sensitivity did not converge and retains its research status. The 50-jitter diagnostic recovered the best solution 48 times and found a poorer mode twice; the age-3+ natural-mortality profile is shallow.

Both pathway-oriented covariate suites also produced OK fits with positive-definite Hessians. Within the physical suite, the combined sea-ice and April–May SST model had the lowest AIC and the largest reduction in recruitment-process variance, but the correlated covariates produced a suppression pattern that limits interpretation of their individual paths. Within the overlap suite, adult-pollock–juvenile-pollock overlap had the lowest AIC; adding arrowtooth overlap provided essentially no further variance reduction. The overlap results remain contemporaneous associations, and the adult-pollock index is partly population-derived, so neither result is treated as a causal recruitment effect.

Immediate work should retain the ADMB-to-Rceattle comparison as a repeatable numerical check, evaluate selectivity and survey influence on the retrospective pattern, and continue sensitivity checks for the shared ATS/AVO annual selectivity deviations. Given acceptance in other applications and the proven bridging aspects, we feel this approach should be adopted for operational management advice for November/December 2026. Future refinements will explore trade-offs in fitting data in alternative ways (e.g., using random effects approaches for selectivity variability, selecting alternatives to the robust-multinomial approach used in this example). The exploratory DSEM work provides a reproducible approach for adding cohort-aligned covariates, while remaining separate from the final assessment configuration and projection results.

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Appendices

A-1 Reproducibility and analysis sequence

The working paper is rendered from saved model and diagnostic products so that building the document does not repeat long model fits. The commands below are run from the project directory, in the order shown when an earlier product is required by a later analysis.

1. The [input-preparation script](#) documents the transformations used to prepare the aligned Rceattle workbook; its precursor workbook is regenerated with Rscript "2024 EBS pollock setup data.R". The frozen workbook used for the reported fit is Data/EBS_24_pollock_m23_rceattle_full_1964–2024.xlsx. The same script documents the modified ADMB reference rebuild, run as `cd ADMB/m23_rceattle_full && admb pm && ./pm -nox -iprint 150` after the ADMB executable directory has been added to PATH.
2. The [two-stage fitting script](#) fits the reported Rceattle configuration and the 2D age-by-year AR1 sensitivity with Rscript "2024 EBS pollock.R". The [5.8.1 summary script](#) then generates the fitted-model summaries and standard-multinomial sensitivity using Rscript R/finalize_rceattle_5_8_1_analysis.R.

3. The [ADMB bridge runner](#) sets the Rceattle parameters and realized selectivity to the ADMB values and saves the trajectory, likelihood, and gradient comparisons with Rscript `R/run_canonical_pm_forward_bridge.R`.
4. The [OSA runner](#) generates the saved OSA residuals with Rscript `R/run_canonical_pm_diagnostics.R`. The [retrospective script](#) runs the nine terminal-year-consistent peels with Rscript `"2024 EBS pollock two-stage retrospectives.R"`.
5. The [5.8.1 self-test runner](#) generates 50 phased simulations with Rscript `R/run_rceattle_5_8_1_self_test.R`. The [restart runner](#) repeats the same simulations from the estimated solution with Rscript `R/run_rceattle_5_8_1_self_test_restart.R`, and the [validation summarizer](#) is run with Rscript `R/summarize_rceattle_5_8_1_validation.R`.
6. The [projection runner](#), which uses the [Rceattle-to-SPM input writer](#), generates the seven-scenario and fixed-catch projection products with Rscript `R/run_spmr_tier3_projection.R`.
7. The [DSEM script](#) runs the four observed-covariate models with Rscript `"2024 EBS pollock DSEM proof of concept.R"` and the missingness-preserving randomized controls with Rscript `"2024 EBS pollock DSEM proof of concept.R" --randomized`. The [separate EcoState-candidate runner](#) is run with `--suite=physical` and `--suite=overlap`; append `--randomized` to each command to reproduce the corresponding randomized controls.
8. After these saved products are available, the HTML and PDF are generated together with `quarto render ebswp.qmd`.

R version 4.6.1 (2026-06-24)

Platform: aarch64-apple-darwin23

Running under: macOS Tahoe 26.6.2

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.

locale:

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time zone: America/Los_Angeles

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] tidyr_1.3.2 tibble_3.3.1 scales_1.4.0 readr_2.2.0
 [5] Rceattle_5.8.1 knitr_1.51 gt_1.3.0 ggthemes_5.2.0
 [9] ggplot2_4.0.3 ggridges_0.5.7 dplyr_1.2.1 asar_2.5.0.9000
 [13] afscOSA_0.0.1

loaded via namespace only:

[1] generics_0.1.4 xml2_1.6.0 stringi_1.8.7 lattice_0.22-9
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 [9] grid_4.6.1 RColorBrewer_1.1-3 fastmap_1.2.0 plyr_1.8.9
 [13] jsonlite_2.0.0 Matrix_1.7-5 tinytex_0.60 purrr_1.2.2
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 [37] fs_2.1.0 vroom_1.7.1 pkgconfig_2.0.3 pillar_1.11.1

Table 21: Files used or referenced by this working paper.

Role	Location	Available	Size
Prepared model-input workbook	Local workbook	TRUE	135.1 KB
BTS covariance input	Local data file	TRUE	12.7 KB
Saved fitted Rceattle objects	Local R serialized-data (RDS) file	TRUE	88014.2 KB
OSA residuals	Local RDS file	TRUE	91.2 KB
OSA diagnostics	Local RDS file	TRUE	1.1 KB
Nine-peel Rceattle retrospective	Local RDS file	TRUE	368646.6 KB
Nine-peel Mohn's rho summary	Local CSV file	TRUE	0.2 KB
Nine-peel retrospective script	Local R script	TRUE	9.7 KB
ADMB-to-Rceattle comparison output	Local RDS file	TRUE	650.8 KB
Rceattle 5.8.1 self-test summary	Local CSV file	TRUE	0.5 KB
Rceattle 5.8.1 truncation diagnostics	Local CSV file	TRUE	0.4 KB
Rceattle 5.8.1 restart comparison	Local CSV file	TRUE	3.8 KB
Rceattle 5.8.1 jitter summary	Local CSV file	TRUE	2.3 KB
Age-3+ natural-mortality profile	Local CSV file	TRUE	0.3 KB
ADMB-to-Rceattle comparison summary	Local comma-separated-values (CSV) file	TRUE	1.0 KB
Rceattle fitting script	Local R script	TRUE	15.7 KB
Diagnostics script	Local R script	TRUE	4.9 KB
Next-assessment update script	Local R script	TRUE	2.6 KB
ADMB comparison script	Local R script	TRUE	10.1 KB
Rceattle-to-SPM input writer	Local R script	TRUE	8.7 KB
spmR Tier 3 projection runner	Local R script	TRUE	1.1 KB
SPM projection manifest	Local CSV file	TRUE	0.5 KB
Exploratory DSEM script	Local R script	TRUE	16.8 KB
DSEM model comparison	Local CSV file	TRUE	0.5 KB
DSEM recruitment-path estimates	Local CSV file	TRUE	3.1 KB
Randomized DSEM control	Local CSV file	TRUE	0.5 KB
Physical and overlap DSEM script	Local R script	TRUE	21.6 KB
Physical-forcing DSEM comparison	Local CSV file	TRUE	0.5 KB
Top-down-overlap DSEM comparison	Local CSV file	TRUE	0.5 KB

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[45]	tidyselect_1.2.1	farver_2.1.2	htmltools_0.5.9	labeling_0.4.3
[49]	rmarkdown_2.31	TMB_1.9.21	compiler_4.6.1	S7_0.2.2
[53]	markdown_2.0			

A-2 Rceattle-to-spmR projection bridge

The projection bridge is a deterministic file adapter rather than a second assessment fit. The [input writer](#) reads the saved `nonparametric_pm` Rceattle object, extracts the population state, biological schedules, fishery schedule, and historical series needed by the Standard Projection Model (SPM), and writes them in the SPM `pm.prj` format. It also writes the projection controls in `spm.dat`. The [projection runner](#) passes those files to `spmR::runSPM(..., engine = "admb")`, which executes the validated SPM binary and returns the simulation-level output used by the report.

The adapter preserves the terminal assessment state and makes each conversion explicit (Table 22). Numbers at age, natural mortality, and the weight schedules are passed on their Rceattle scales. With abundance in millions of fish and weight in kilograms, their product has units of thousand metric tons. Maturity is divided by its maximum, matching the SPM input convention used for this stock. The spawning-month encodings differ by one because Rceattle records elapsed months and SPM uses a one-based calendar month; both represent survival through 0.25 of the year.

Table 22: Mapping from the saved Rceattle fit to the SPM species input file.

SPM field	Rceattle source	Transformation and years
Average recent F	quantities: F_spp	Arithmetic mean for 2020–2024; used by Alternative 3
Natural mortality at age	quantities: M_at_age	Terminal-year schedule (2024)
Maturity at age	data_list: maturity	Divided by its maximum to match the SPM convention
Spawning weight at age	data_list: weight and ssb_wt_index	Terminal-year spawning-weight schedule
Fishery weight at age	data_list: weight, fishery index	Terminal-year fishery-weight schedule
Fishery selectivity at age	quantities: sel_at_age for Fishery	Arithmetic mean at each age for 2020–2024
Terminal numbers at age	quantities: N_at_age	Terminal assessment state (2024)
Recruitment history	quantities: R	Annual age-1 recruitment for 1978–2024
Spawning-biomass history	quantities: ssb	Annual spawning biomass for 1977–2023
Spawning month	data_list: spawn_month = 3	Written as SPM Spawnmo = 4; both equal 0.25 year

Annual Rceattle fishery selectivity is already normalized to a mean of one across ages. The projection schedule is calculated as the arithmetic mean at each age across 2020–2024; averaging these normalized annual curves retains a mean of one. This five-year schedule replaces the former terminal-year-only curve. The writer saves the five contributing curves in `fishery_selectivity_recent_years.csv`, the applied average

in `age_schedules.csv`, and the rule and exact years in `lineage.csv`. Thus the projection input can be reconstructed without reading values back from the formatted `pm.prj` file.

The `spm.dat` file starts projections in 2025, runs 14 projection years and 1,000 simulations, and requests all seven Tier 3 alternatives. Catch is fixed at 1,350 thousand t in 2025 and 2026 before the alternatives diverge. A separate call uses Alternative 2 with catch fixed at 1,300 thousand t for 2025–2032. The writer copies the validated SPM executable and `tacpar.dat` control file into each run directory. `manifest.csv` records the role, presence, and MD5 checksum of every required input, while `lineage.csv` records the Rceattle source checksum, terminal year, Rceattle and spmR versions, selectivity rule, and averaging years.

The saved products passed the bridge checks in Table 23. These checks establish file and field consistency; they do not replace review of the harvest assumptions or the biological appropriateness of a five-year selectivity average.

Table 23: Automated consistency checks for the Rceattle-to-spmR projection bridge.

Check	Result	Evidence
Required projection files	Pass	6 of 6 manifest targets present
Recent selectivity years	Pass	2020, 2021, 2022, 2023, 2024
Applied five-year selectivity mean	Pass	maximum absolute difference = 3.996803e-15
Projection alternatives	Pass	1, 2, 3, 4, 5, 6, 7
Projection start year	Pass	2025

A-3 Model fitting details

The fishery selectivity likelihood has multiple modes, so the reported Rceattle configuration used data-based fishery-selectivity starting values and two fitting stages. The first stage disabled the annual selectivity deviations to establish the selectivity and abundance scale. The second stage restored the deviations and used the first-stage solution as its initial values. This sequence produced a small maximum gradient and OK status. With the direct ATS/AVO coefficient-increment penalty, the Hessian condition number was approximately 489,000; its weakest direction remained in the shared ATS/AVO annual selectivity deviations (`sel_coff_dev`) rather than fishery/CPUE.

The complete fitting sequence, including the empirical starting values and the two estimation stages, is in the [two-stage fitting script](#) and is run with `Rscript "2024 EBS pollock.R"`. The nine-peel implementation is in the [retrospective script](#) and is run with `Rscript "2024 EBS pollock two-stage retrospectives.R"`.

The retrospective peels repeat this sequence after recalculating the data-based selectivity starting values for each terminal year. Because `sel_coff_dev` contains annual random-walk increments at age, each peel fixes the terminal-year fishery and mirrored catch-per-unit-effort increments at zero. This holds their terminal selectivity-at-age curves at the preceding-year curves.

A-4 Jitter and natural-mortality profile

The [extended-diagnostics runner](#) generates both the 50 jitter fits and the age-3-and-older natural-mortality profile with `Rscript R/run_rceattle_5_8_1_extended_diagnostics.R`.

The 50 phased jitter runs test sensitivity to dispersed starting values. Forty-eight returned to the best objective within numerical precision. Two received WARN status and reached an inferior mode with an objective increase of approximately 609.5. This reproduces the pre-existing multimodality and shows why convergence of one run is not sufficient evidence that the global mode was found.

The age-3-and-older natural-mortality profile was evaluated from 0.15 to 0.50. The grid minimum occurred at 0.20, but the objective changed by only 3.82 across the entire grid. The profile is therefore shallow and does not support precise estimation of natural mortality from these data alone.

Table 24: Disposition of the Rceattle development components.

Item	Development component	Review result	Interpretation
1	Prepared Rceattle model-input workbook	Used the delivered workbook directly.	Input files are available and reproducible from stored files.
2	Rceattle fit and 2D AR1 sensitivity	The Rceattle fit completed; 2D AR1 received a FAIL convergence status.	The NonParametricPM configuration is the relevant Rceattle comparison; 2D AR1 is diagnostic only.
3	Convergence and model diagnostics	Structured convergence checks, OSA diagnostics, nine retrospective peels, 50 self-tests, 50 jitters, and a natural-mortality profile are saved.	The ATS/AVO coefficient-increment penalty resolves the prior Hessian-conditioning warning; two inferior jitter solutions remain.
4	Next-assessment data update workflow	Update script review is complete; execution awaits complete new-year observations.	Useful as a template only after complete new-year observations are inserted.
5	ADMB-to-Rceattle comparison diagnostics	Setting Rceattle to the ADMB values reproduces ADMB to rounding-scale precision.	This is the strongest evidence that the ADMB comparison is correct.

Table 25: Rceattle 5.8.1 phased-jitter results for the fitted configuration. Delta objective is measured from the best of 50 runs.

Status	Runs	Minimum delta objective	Maximum delta objective
OK	48	0.000	0.000
WARN	2	609.548	609.548

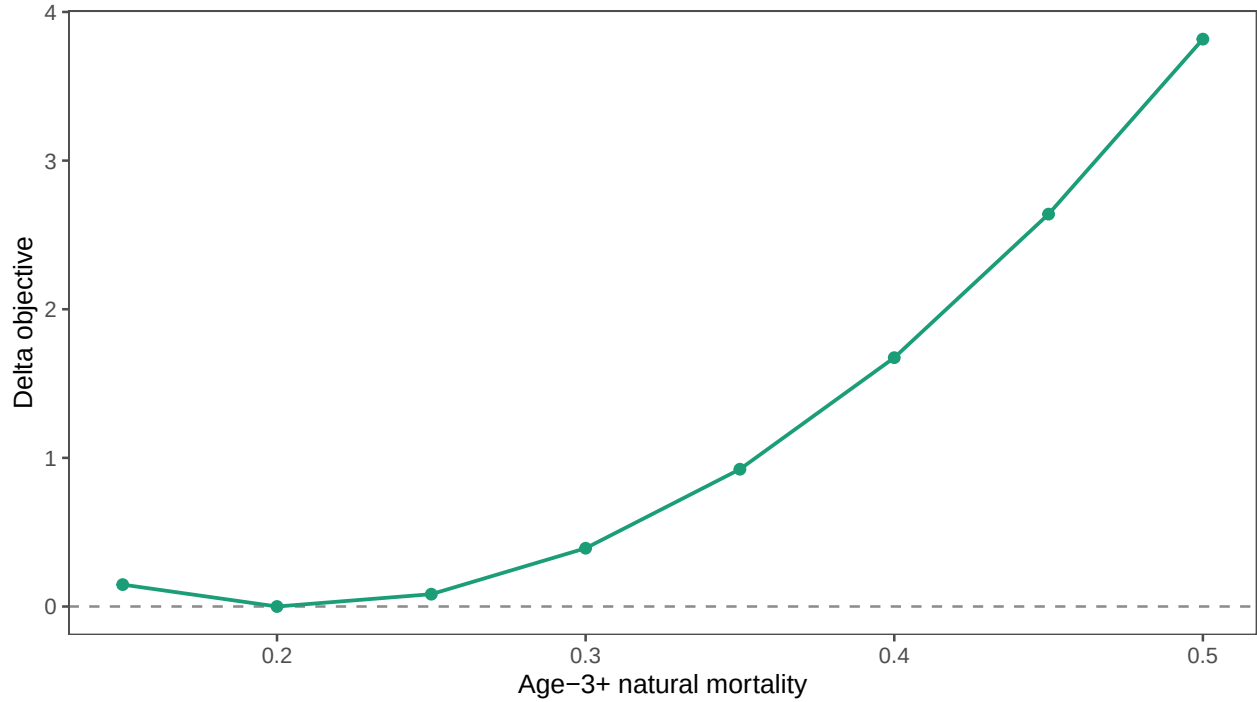


Figure 23: Profile of age-3-and-older natural mortality for the Rceattle 5.8.1 configuration. Points are evaluated grid values and the vertical axis is objective difference from the grid minimum.

Table 26: Composition-likelihood sensitivity for the Rceattle 5.8.1 configuration. Raw objectives are shown for run provenance but are not compared because likelihood constants and conventions differ.

Model	Composition likelihood	Objective	Maximum gradient	Fit status
Final configuration	AFSC multinomial	716.948	4.14×10^{-5}	OK
Standard multinomial	Standard multinomial	2,843.416	2.80×10^{-4}	OK

A-5 Composition-likelihood sensitivity

The [5.8.1 summary script](#) fits the standard-multinomial sensitivity and writes the summaries used here when run with `Rscript R/finalize_rceattle_5_8_1_analysis.R`.

The Rceattle configuration uses the supported `MultinomialAFSC` composition likelihood with a robustness constant of 0.001 added to observed and expected proportions. A standard multinomial fit without this constant was retained as an observation-model sensitivity. Both fits reached small gradients and received OK status. The final configuration’s residual least-identified direction remained in the shared ATS/AVO annual selectivity deviations.

Relative to the `MultinomialAFSC` configuration, the standard multinomial sensitivity changed spawning biomass by 1.79%, recruitment by 1.05%, and total biomass by 11.09% on average in absolute percentage terms.

Table 27: Diagnostic 2D AR1 fishery-selectivity correlation and scale estimates.

Quantity	Rceattle parameter	Penalized effects	Random-effects fit
Year AR1 correlation	Sel_curve_pen1	0.584	0.947
Age AR1 correlation	Sel_curve_pen2	0.120	0.098
Selectivity deviation scale	sel_dev_log_sd	0.500	0.379

Correlations are on the -1 to 1 scale; the deviation scale is a positive standard deviation. The fit failed convergence checks.

A-6 2D age-by-year AR1 selectivity sensitivity

The penalized-effects initialization and subsequent Laplace fit are in the [two-stage fitting script](#) and are run with `Rscript "2024 EBS pollock.R"`. A separate [fixed-scale follow-up](#) can be run with `Rscript "2024 EBS pollock 2D AR1 fixed sd test.R"`; it is a development check and is not the sensitivity summarized below.

The 2D AR1 sensitivity applied an age-by-year field to the shared fishery and catch-per-unit-effort selectivity block over 1964–2024. The age-year deviations were first fit as penalized effects to initialize the field and then integrated as random effects with a Laplace approximation. The initialized Laplace fit failed the gradient, Hessian, and bound checks.

The failed random-effects fit had a maximum marginal gradient of 19.7 on `rec_pars`, a non-invertible Hessian, 61 `log_F` parameters at their configured bounds, and **FAIL** convergence status. These estimates remain diagnostic.

A-7 Acronyms

This appendix follows the glossary convention in the [Alaska Stock Assessment Report \(asar\)](#) package (Schiano et al. 2026). The project glossary uses ASAR’s `\newacronym` format and adds terms specific to this Rceattle comparison. The quoted label “PM” is a model name.

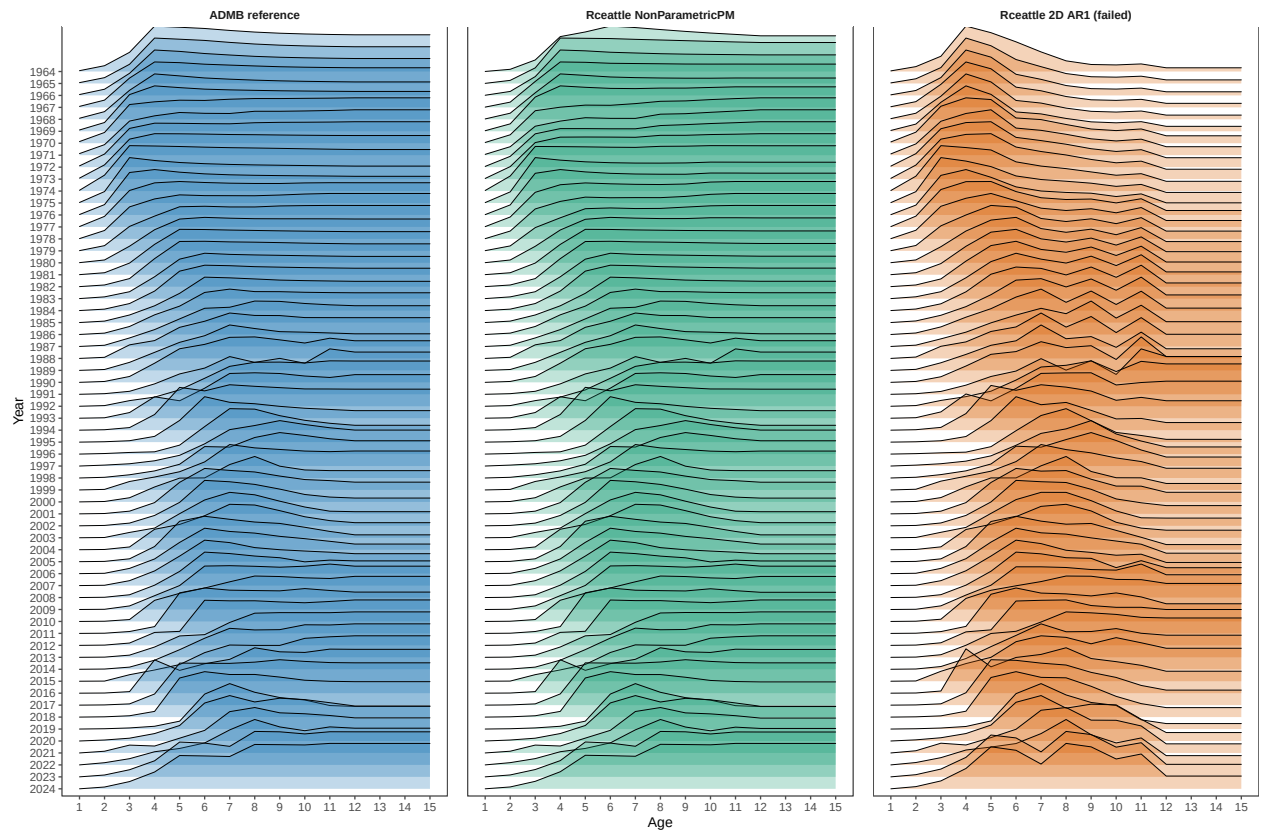


Figure 24: Development comparison of relative fishery selectivity for ADMB, the Rceattle configuration, and the failed 2D age-by-year AR1 sensitivity.

Table 28: Acronyms used in this working paper.

Acronym	Definition
ADMB	AD Model Builder
AIC	Akaike information criterion
AR1	first-order autoregressive
ATS	acoustic-trawl survey
AVO	acoustic vessels of opportunity
BTS	bottom trawl survey
CIE	Center for Independent Experts
CPUE	catch per unit effort
CSV	comma-separated values
DAG	directed acyclic graph
DSEM	dynamic structural equation modeling
ESP	ecosystem and socioeconomic profile
HTML	HyperText Markup Language
IID	independent and identically distributed
MCMC	Markov chain Monte Carlo
MVN	multivariate normal
OSA	one-step-ahead
RDS	R serialized data
RTMB	R Template Model Builder
SAFE	Stock Assessment and Fishery Evaluation
SDNR	standard deviation of normalized residuals
SEM	structural equation model
SPM	Standard Projection Model
SST	sea surface temperature